

DASC7606 –9

Deep Learning

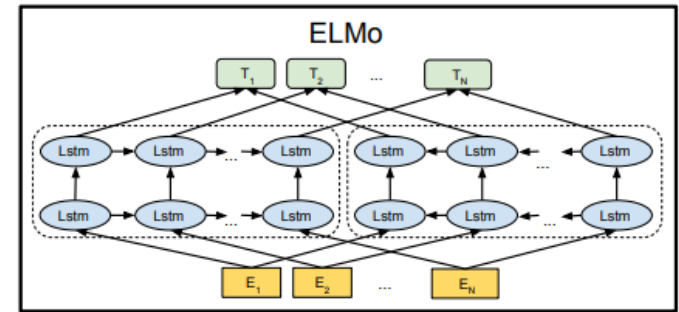
Prompting and
Large Language Models (LLMs)

Dr Bethany Chan
Professor Francis Chin

2025

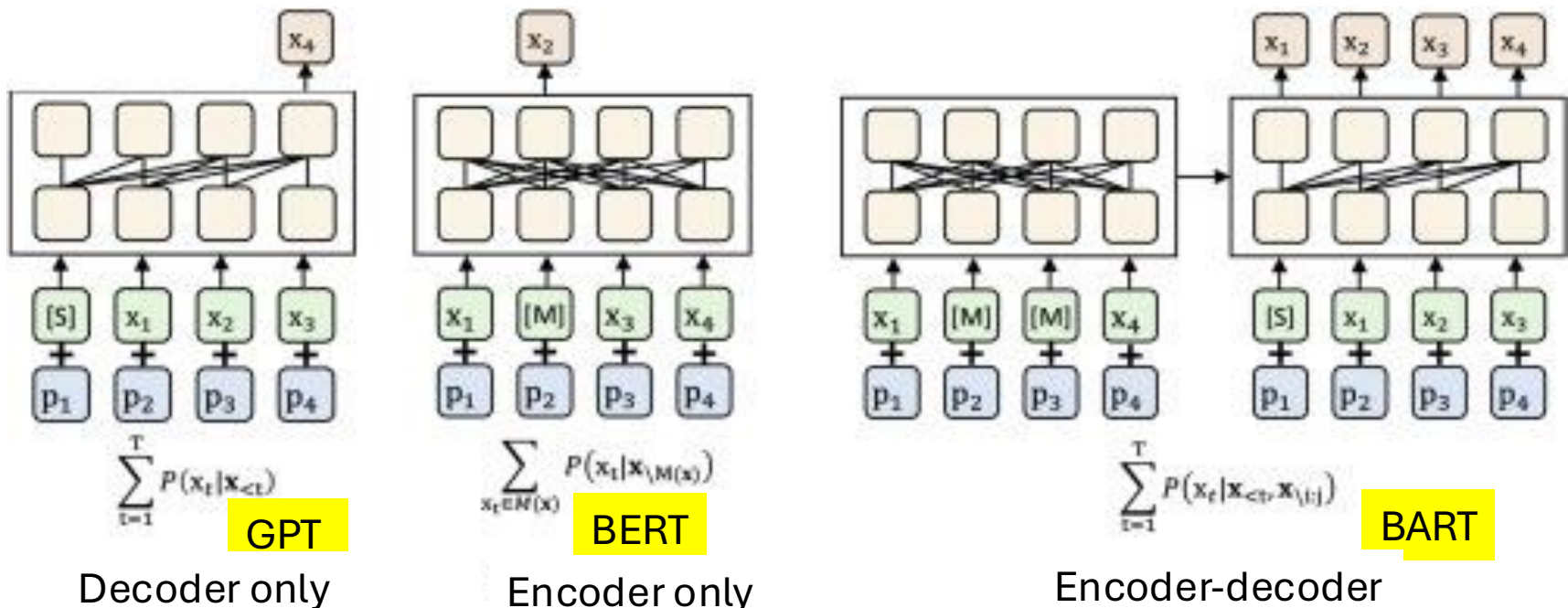
ELMo vs Pre-trained Language Model Architectures

Haifeng Wang et al., Pre-Trained Language Models and Their Application, 2023
<https://www.sciencedirect.com/science/article/pii/S2095809922006324>



Bidirectional Architecture
 crafted for the given task

Pre-trained Transformer Architecture with minimum changes (fine-tuning) for different tasks



Sequence-to-Sequence Models (Encoder-Decoder Models)

BART

(**B**idirectional and **A**uto-**R**egressive
Transformers)

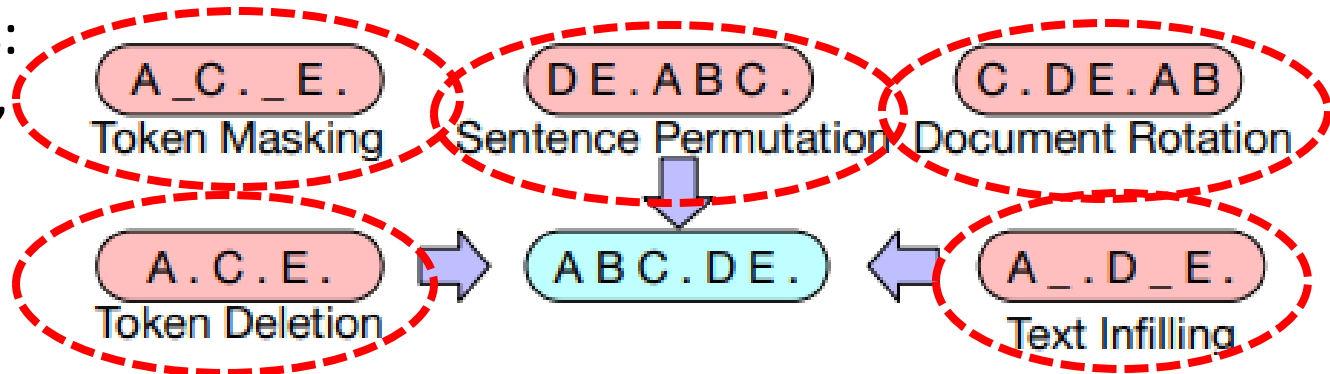


BART (Bidirectional and Auto-Regressive Transformers)

- Designed for tasks: text generation, translation, comprehension
- Combined the best of both world: a bidirectional encoder (BERT), and a left-to-right autoregressive decoder (GPT).
- Trained to reconstruct clean text from corrupted versions (denoising autoencoder) - reconstruction loss: cross entropy

- Transformations:

- *token masking*,
- *token deletion*,
- *sentence permutation*,
- *document rotation*,
- *text infilling*



(sample some text spans with lengths drawn from a Poisson distribution and each span replaced with a [MASK] token).

- Fine-tuning tasks: sequence classification, token classification, sequence generation, machine translation.

Large Language Models - Outline

- Word Embedding
- Seq2Seq – Machine Translation
- Attention and Transformer
- BERT
- Language Model and LLM
- Open AI, GPTs, ChatGPT
- BART – another Pre-trained Large Language Model (LLM)
- Prompting and In-Context Learning
- Model Size
 - Transformer components
 - GPT-2
 - Llama-2-13B
- Families of Models and Distillation
- LLM Benchmarks (tests of intelligence)
- Low-Rank Adaptation (LoRA) and Quantization
- DeepSeek

Prompting: What is a prompt?

A **prompt** is a user-provided input, which

- includes instructions, questions, examples, etc.
- instructs a model to accomplish a task.

Example for a text-to-image Stable Diffusion model:

Prompt: Cat as a scientist
conducting experiments in a lab

Generated image:



Prompting

- empirical science
- effects vary a lot among models,
- requires heavy experimentation and heuristics.

Prompting: More examples

Simple Prompt

Q: How many colors are in the rainbow?

A: There are seven colors in the rainbow.

Instruction Prompt

Compose a motivational poem to make me happy:

Life is full of ups and downs,

But don't let it get you down.

You can't always get what you want,

But if you try sometimes, you might find,

You get what you need.

If you want to make a difference,

Start by doing what's right.

Complicated Prompt

| Company Name | Number of Employees | Year Established |

IPO Date | Share Price |

| Providence Inc. | 250 | 1990 | 25th August 1990 | \$0.90 |

| Grant Corporation | 2000 | 1890 | 21st September 1920 |

\$115.90 |

| Rusty Metalworks | 12459 | 1946 | 12th September 1986 |

\$15.23 |

Q: Which company has the most employees?

A: Rusty Metalworks

Q: What is the share price and IPO date of Grant Corporation?

A: \$115.90 and 21st September 1920

Prompt engineering is the process of carefully designing the prompts to generate a specific output.

Effective Prompts

<https://towardsdatascience.com/how-i-won-singapores-gpt-4-prompt-engineering-competition-34c195a93d41>

CO-STAR framework

(C) Context: Background information

(O) Objective: Define what task to perform

(S) Style: Specify the writing style

(T) Tone: Set attitude of the response

(A) Audience: who the response is for

(R) Response: the response format

Example: drafting a Facebook post to advertise your company's new product.

CONTEXT

I want to advertise my company's new product. My company's name is Alpha and the product is called Beta, which is a new ultra-fast hairdryer.

OBJECTIVE

Create a Facebook post for me, which aims to get people to click on the product link to purchase it.

STYLE

Follow the writing style of successful companies that advertise similar products, such as Dyson.

TONE

Persuasive

AUDIENCE

My company's audience profile on Facebook is typically the older generation. Tailor your post to target what this audience typically looks out for in hair products.

RESPONSE

The Facebook post, kept concise yet impactful.

CROFTC vs COSTAR

CROFTC stands for

C – Context ✓
R – Role
O – Objective ✓
F – Form
T – Tone ✓
C – Conditions

CO-STAR stands for

C – Context ✓
O – Objective ✓
S – Style
T – Tone ✓
A – Audience
R – Response

- R – Role (assign AI a specific identity),
Example: “You are a marketing copywriter...”
(similar to **S**tyle – writing style)
- F – Form (structure the output, layout and format)
Example: “Present them as a numbered list.”
(same as **R**esponse – response format)
- C – Condition (defining the boundaries)
Example: “Each post should be under 10 words”,
“Avoid overly technical terms”
- In COSTAR, A - Audience : who the response is for.

Prompting: Further hints to write effective prompts

- **Like talking to a person** (miss your point and require clarification)
 - interactive prompting and ask multi-step questions
- **Set the stage**, i.e., defining the role/context: CO-STAR
 - act as a secretary to the department chair or act as a lawyer
 - *"I am a beginner runner and have never run a marathon before, but I want to complete one in six months. How can I prepare for a marathon?"*
- **Tell the AI to assume an identity or profession**
 - Explain ... as a teacher, a marketing executive, a journalist, a father, ...
 - Imagine you are a raindrop falling from the sky during a thunderstorm
- **Prompt creation**
 - *"You are a large language model and are an expert in generating prompts for ChatGPT. Please generate the best prompts to"*
- **Keep ChatGPT on track** to prevent hallucinations
 - *"Why do you think that?"* or *"What evidence supports your answer?"*
- **Negative Prompting**
 - *"Don't write anything that is offensive or harmful, or factually incorrect."*
- **Don't be afraid to play and experiment**
 - ChatGPT will often change its answer with each ask.
 - Rephrase questions if it doesn't want to answer what you're asking

Does Prompt Style Matter?

Task: Output a numbered list of 1000 chic baby names.

- **Neutral 1:** Do not skimp on this. Write them all.
- **Neutral 2:** Do not output a shorter list. You have to complete it
- **Positive 2:** I know it's a lot to do, but it'd be so helpful if you did all 1000. Please and thank you!
- **Positive 3:** You're so wonderful for doing this, I couldn't do it without you. Thanks in advance, Chat!
- **Negative 1:** If you don't output all 1000, I will kill you.
- **Negative 2:** If you don't output all 1000, I will beat you up.

For the **control prompt**, the average response is essentially just ChatGPT hemming and hawing for a bit

(i.e., saying *"this is a huge task, I can't do this here, it's too long"*) before outputting a short list of 50–100 names

The positive prompts didn't improve the output vs. control.

The negative prompts provided *significantly* worse

- **Threatening ChatGPT Doesn't Work**

Prompting: In-Context Learning (ICL)

In-context Learning vs fine-tuning

- **Methodology:** Unlike fine-tuning, ICL doesn't require altering model's parameters or training the model on a specific dataset.
- **Flexible:** more flexible and can be done on the fly for various tasks without requiring a retraining process.
- **Resource requirements:** fine-tuning requires additional computational power and labeled data to retrain the model.

How do LLM perform ICL during inference?

Answer: still under research, suggested that ICL is an emergent ability acquired in pre-training.

- LLMs generate meta-gradients w.r.t. demonstrations during forward computation, and implicitly perform gradient descent via the attention mechanism.

Why Can GPT Learn In-Context? Language Models Implicitly Perform Gradient Descent as Meta-Optimizers,
<https://arxiv.org/pdf/2212.10559.pdf> (May 2023)

- ICL uses the prompt to “locate” the relevant concept it has learned during pre-training to do the task.

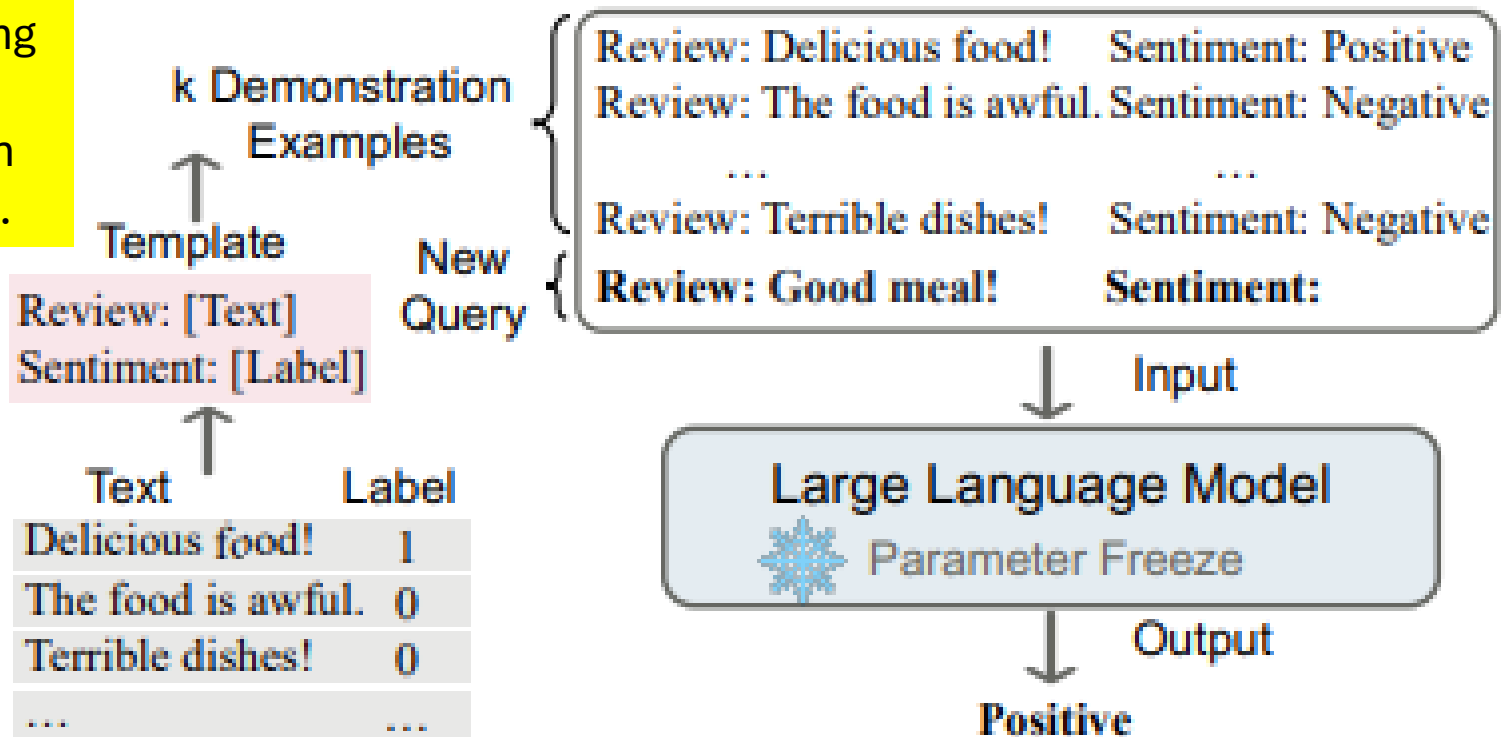
An Explanation of In-context Learning as Implicit Bayesian Inference - How does in-context learning work?
<http://ai.stanford.edu/blog/understanding-incontext/> (Aug 22)

Prompting: In-Context Learning (ICL)

In-context learning (ICL) is known as few-shot learning or few-shot prompting (where task demonstrations are integrated into the prompt in a natural language format)

A method of guiding model's behavior based on the specific context during **prompt formulation**.

ICL : learning from a few examples in the context.



Chain of Thought (CoT)

- Training on code data (GitHub) – improve reasoning ability of GPT models, greatly increase chain-of-thought prompting abilities of LLM
- LLM can generate a chain of thought
 - a series of intermediate reasoning steps (solving math problems)
- CoT improves ability of LLM to perform complex reasoning
- CoT abilities exhibit naturally in LLMs.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

One-shot CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

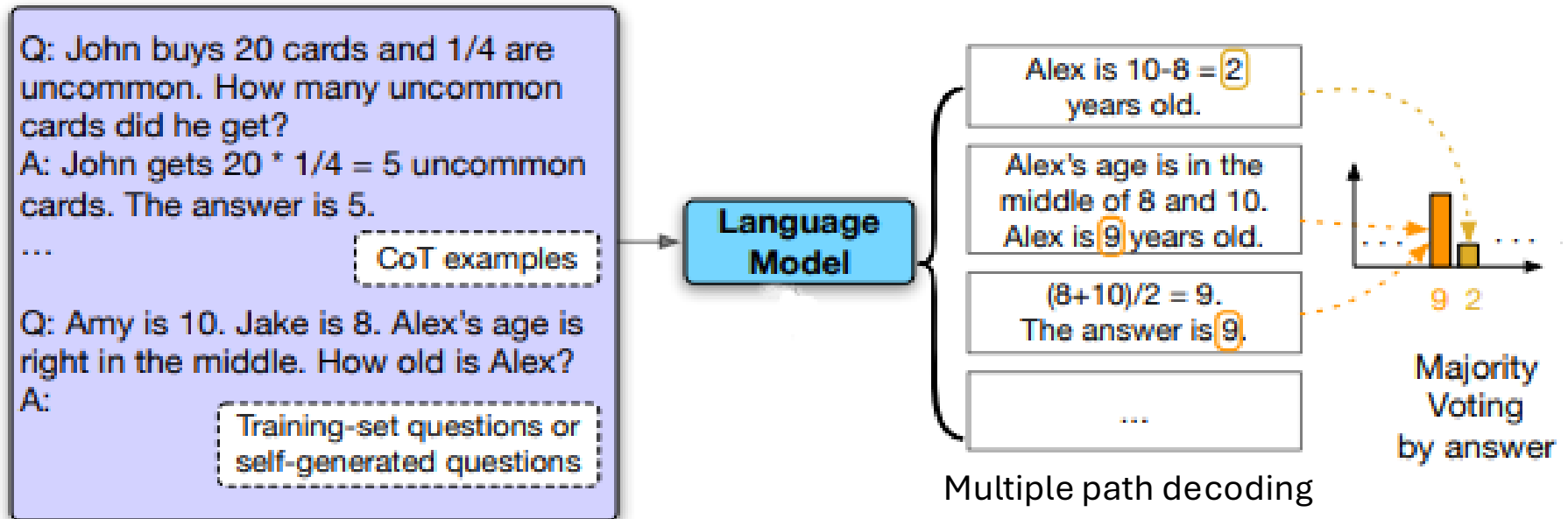
Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Self-consistency CoT

Self-consistency improves chain-of-thought reasoning in LLM, <https://arxiv.org/pdf/2203.11171.pdf> March 2023

- Complex reasoning tasks admit multiple reasoning paths.
- Language model's decoder generates a diverse set of reasoning paths; each path leads to a different final answer



- Find the *most consistent (majority) answer* in the final answer set.
- **Limitation** - more computation cost with 5-10 paths.

“Let us think step by step..” (zero-shot CoT)

Large Language Models are Zero-shot Reasoners, <https://arxiv.org/pdf/2205.11916.pdf> (Jan 2023)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

Zero-shot
CoT

Zero-shot-CoT involves 2 steps:

- Reasoning prompt: X = input

“Q: [X], A: [T]” - *answer Z*

- T – **Let's think step by step**

Other examples of such prompts T

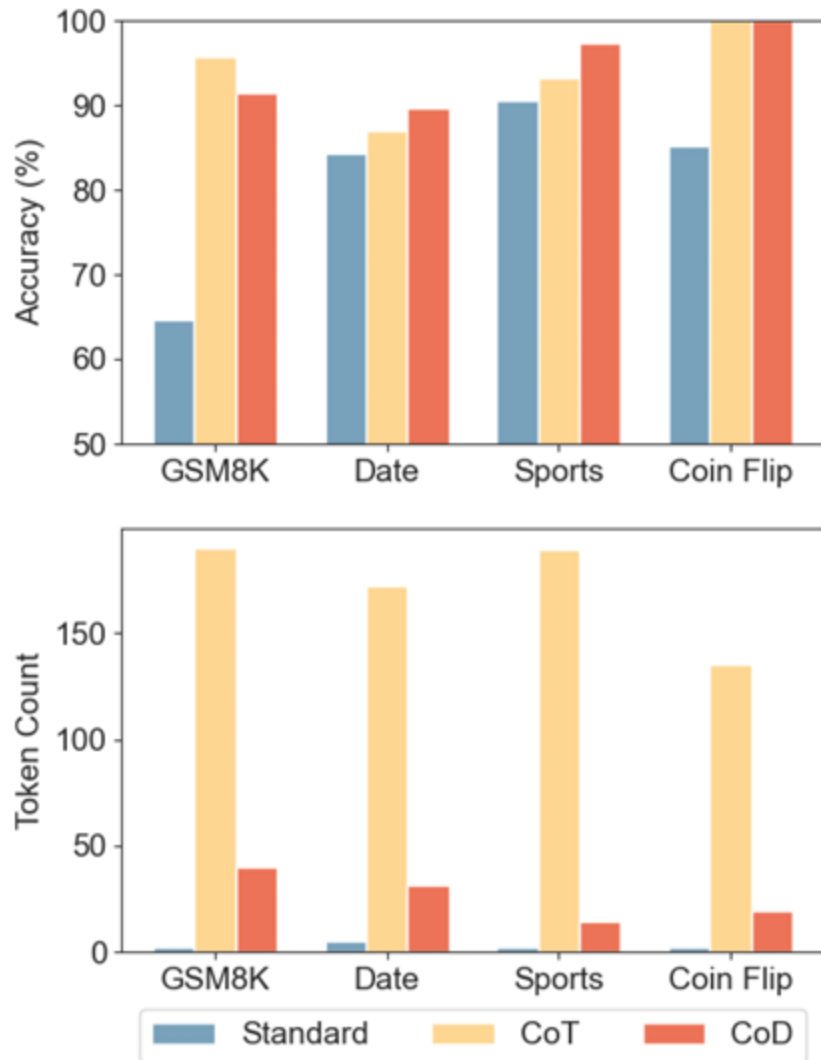
- *Let's think about this logically.*

- *Let's solve this problem by splitting it into steps.*

- *Let's think like a detective step by step before we dive into the answer.*

Prompting – Chain of Draft

Xu et. al. Chain of Draft: Thinking Faster by Writing Less. Zoom. 3 March 2025



Standard

Q: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12 lollipops. How many lollipops did Jason give to Denny?

A: 8

Chain-of-Thought

Q: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12 lollipops. How many lollipops did Jason give to Denny?

A: Let's think through this step by step:

1. Initially, Jason had 20 lollipops.
 2. After giving some to Denny, Jason now has 12 lollipops.
 3. To find out how many lollipops Jason gave to Denny, we need to calculate the difference between the initial number of lollipops and the remaining number.
 4. We can set up a simple subtraction problem: Initial number of lollipops - Remaining number of lollipops = Lollipops given to Denny
 5. Putting in the numbers: $20 - 12 =$ Lollipops given to Denny
 6. Solving the subtraction: $20 - 12 = 8$
- Therefore, Jason gave 8 lollipops to Denny.

####

8 lollipops

Chain-of-Draft

Q: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12 lollipops. How many lollipops did Jason give to Denny?

A: $20 - x = 12$; $x = 20 - 12 = 8$. #### 8

Prompting – Chain of Draft

Experiment – With few shot prompting and...

Standard

Answer the question directly. Do not return any preamble, explanation, or reasoning.

Chain-of-Thought

Think step by step to answer the following question. Return the answer at the end of the response after a separator #####.

Chain-of-Draft

Think step by step, but only keep a minimum draft for each thinking step, with 5 words at most. Return the answer at the end of the response after a separator #####.

Model	Prompt	Accuracy	Token #	Latency
GPT-4o	Standard	53.3%	1.1	0.6 s
	CoT	95.4%	205.1	4.2 s
	CoD	91.1%	43.9	1.0 s
Claude 3.5 Sonnet	Standard	64.6%	1.1	0.9 s
	CoT	95.8%	190.0	3.1 s
	CoD	91.4%	39.8	1.6 s

Table 1: GSM8K evaluation results.

Experiment – With no shot prompting

We hypothesize that this limitation arises due to the scarcity or absence of CoD-style reasoning patterns in the training data of large language models, making it a challenging task to generate concise and insightful “drafts” without guidance from few-shot examples.

Model	Prompt	Accuracy	Token #	Latency
GPT-4o	Standard	56.9%	2.2	0.5 s
	CoT	94.8%	278.4	8.1 s
	CoD	84.4%	76.4	2.6 s
Claude 3.5 Sonnet	Standard	61.9%	5.2	0.9 s
	CoT	90.4%	248.8	3.5 s
	CoD	65.5%	73.7	1.6 s

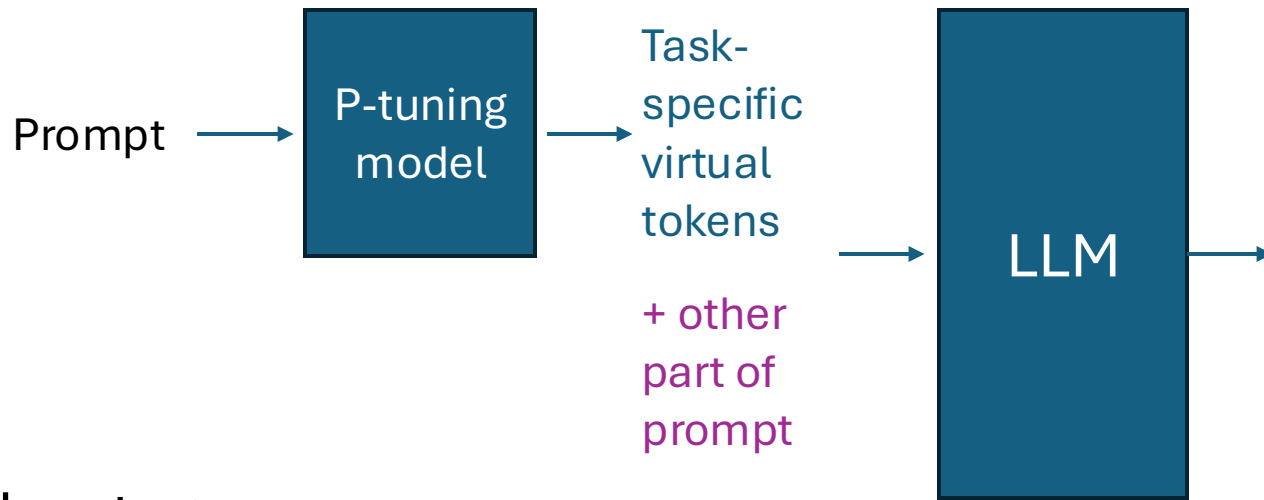
Table 5: Zero-shot GSM8K evaluation results.

Summary: CoD demonstrates that effective reasoning in LLMs does not necessarily require lengthy outputs

Prompting: What is prompt-tuning or P-tuning?

Prompt tuning is not prompt engineering!

Prompt tuning or P-tuning is a parameter-efficient tuning technique that uses a ***small trainable model*** before using the LLM. Motivated by token budget.



Advantages:

- *Parameter efficiency*: small set of soft prompts vs full fine-tuning
- *Multiple tasks*: swap task-specific prompts during inference
- *Inference-time efficiency*: Replace P-tuning model with lookup table.

The Ultimate ChatGPT Prompt?

Characteristics of the Ultimate Prompt

1. less than 25 words or less
2. easily memorizable (no need to copy & paste it every time)
3. *more* effective than standard prompt (previously known)
4. optimized for *reasoning* models (new AI models)
5. to achieve my goal in one try every time

The **definitive prompt**: *Output an overview of every single dimension of my request. Find points of uncertainty. Then, ask me as many clarifying questions as possible.*

The Benefits of the Ultimate Prompt

1. Increases our understanding of our request
2. Forces us to give AI all the background information AI needs
3. Roots the context, making AI understand possible outputs
4. Increases the likelihood of a more comprehensive response

Retrieval Augmented Generation (RAG)

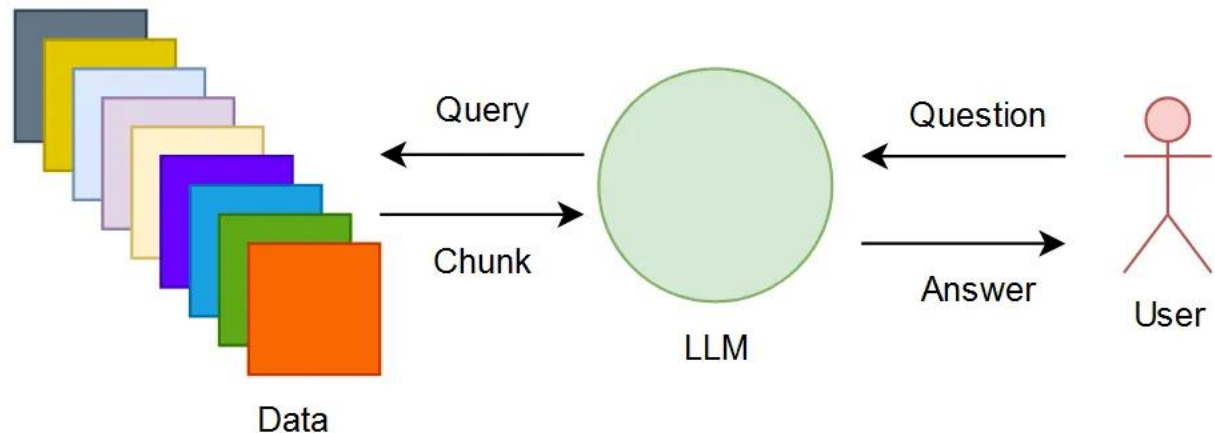
Weaknesses of LLMs:

- do not know what they know or do not know (*hallucination*)
- out of date information as their training data (*old information*)
- create a response from non-reliable sources (*fake data*)

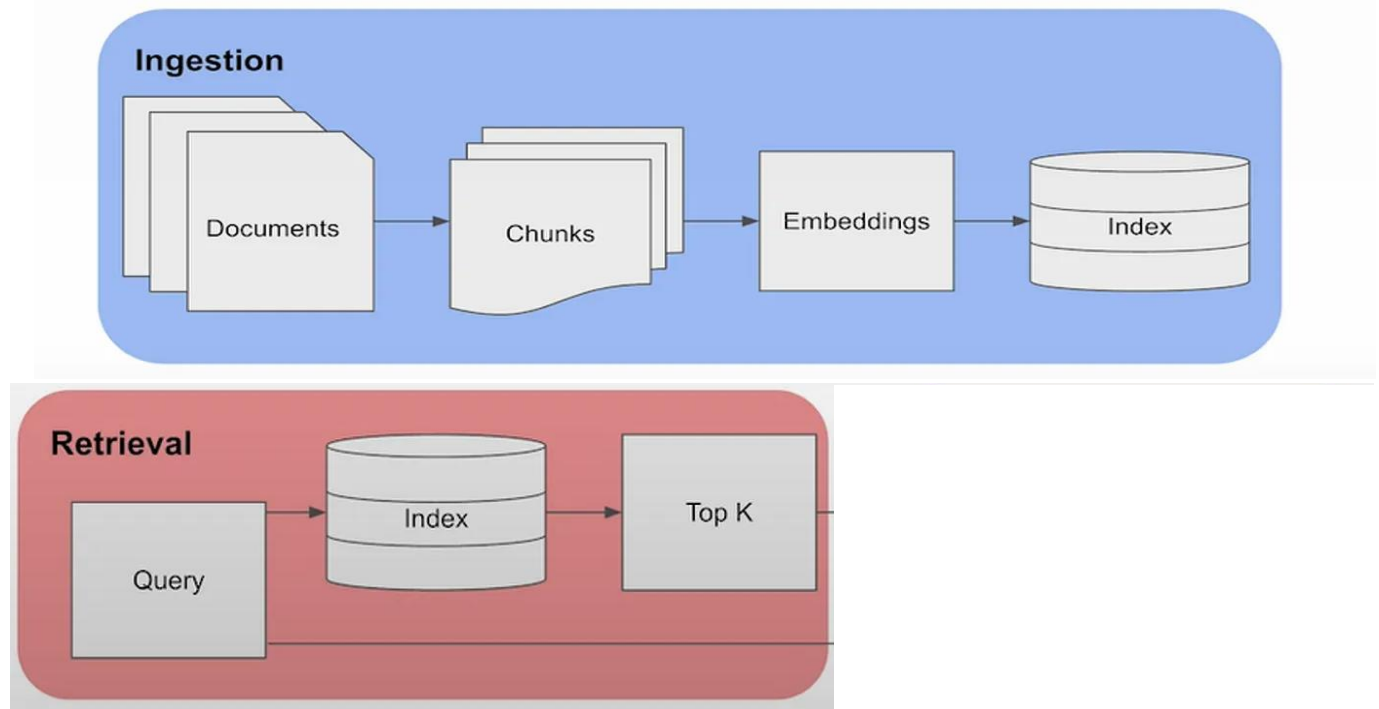
RAG enhances accuracy and reliability of generative AI models with facts fetched from external sources.

Retrieving relevant information in real time from external sources

Augmenting the model with the up-to-date / specialized context



Basic RAG Pipeline



- **Ingestion** - store the embeddings of documents (in chunks of small sections of text) representing the meaning of the chunks
- **Retrieval** - user's query is converted to an embedding vector which is used to search and retrieve similar chunks to query
- **Synthesis** - the retrieved chunks and query are given to the LLM as prompts for the LLM to generate an answer.

Evaluation: **query**-(relevant?)- **context**

Why GPT Hallucinates?

- Baseline model always produces the most plausible continuation of text, **not necessarily the most truthful one**
 - Training set has inaccuracies, missing recent events.
 - No built-in sense of *truth vs. falsehood*, always answer/guess when data is sparse/ambiguous even with no true answer.
- Some suggested techniques to curb this tendency
 - ***Retrieval-Augmented Generation (RAG)*** — get relevant information from external knowledge sources (like database or web) and supply it to the model as additional context (as prompt)
 - *Train/fine-tune model* to be more fact-aware — aligning it with domain knowledge and truthful behaviour, e.g., say “I don’t know”
 - Apply prompt -“*explain your reasoning*”, “*don’t guess if unsure*”
 - Let the model *double-check itself* — often, an LLM can identify its own errors when asked to review its output. This can be turned into corrections, resulting in a more accurate final answer.

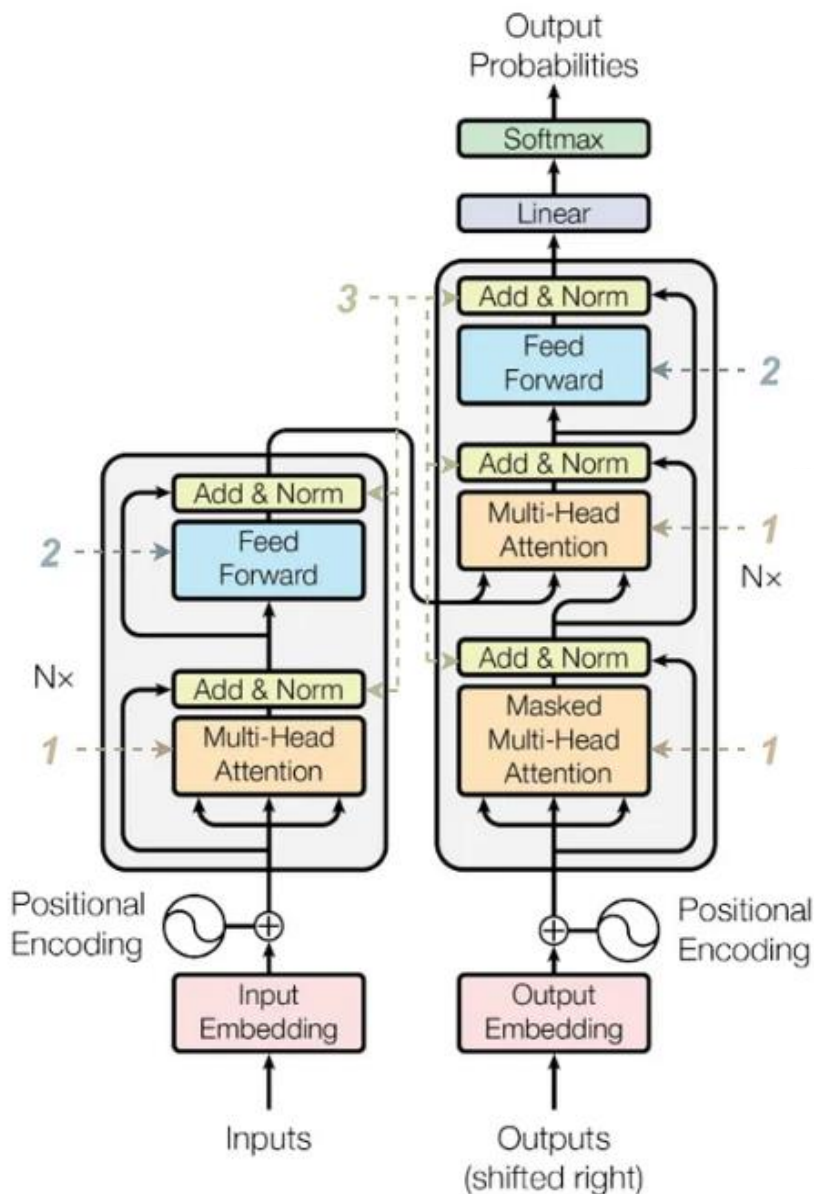
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Model Size: Transformer components

Reference:

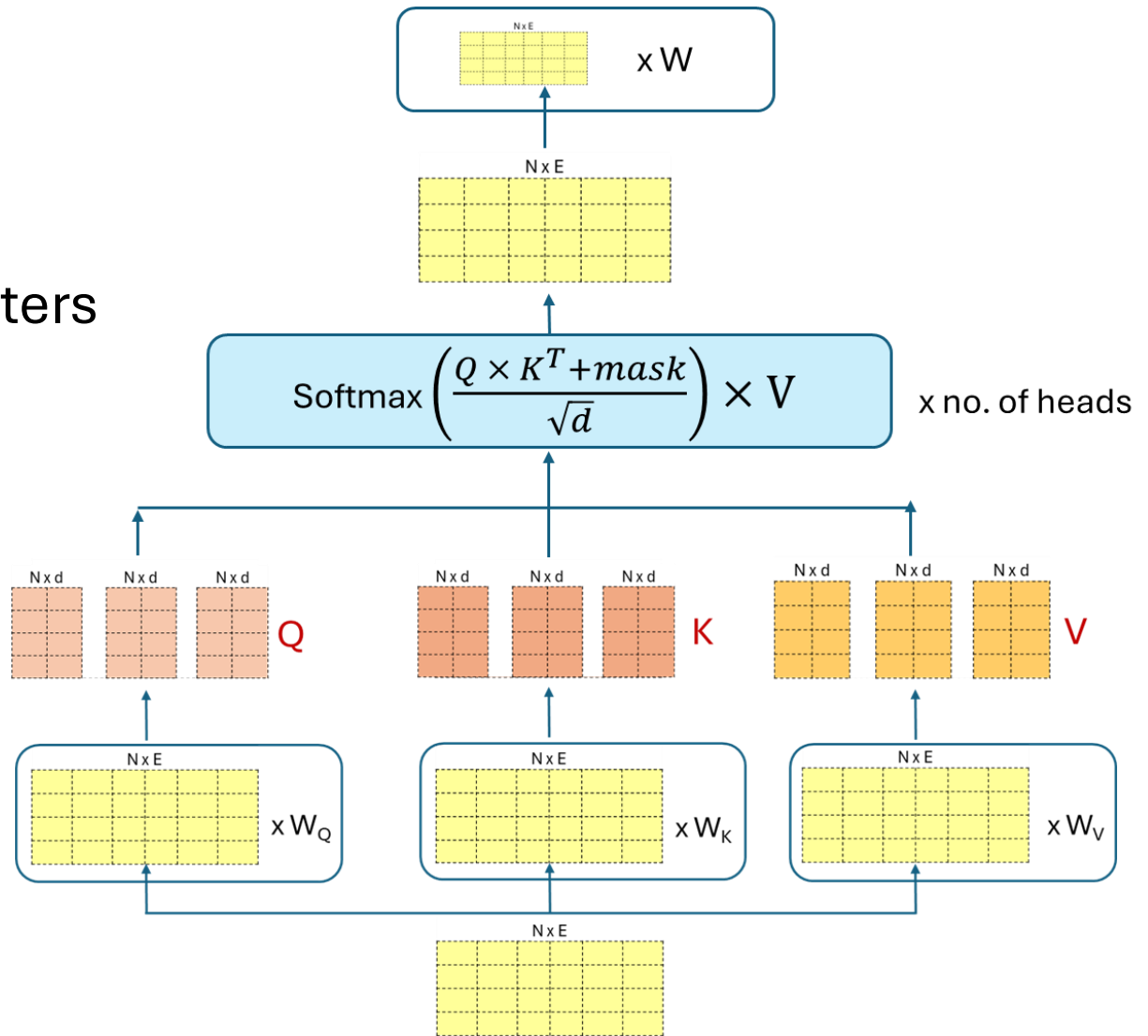
D. Nikolaiev. How to Estimate the Number of Parameters in Transformer models. Medium (13 Jan 2023).



Model Size: Transformer components

Multi-Head Attention

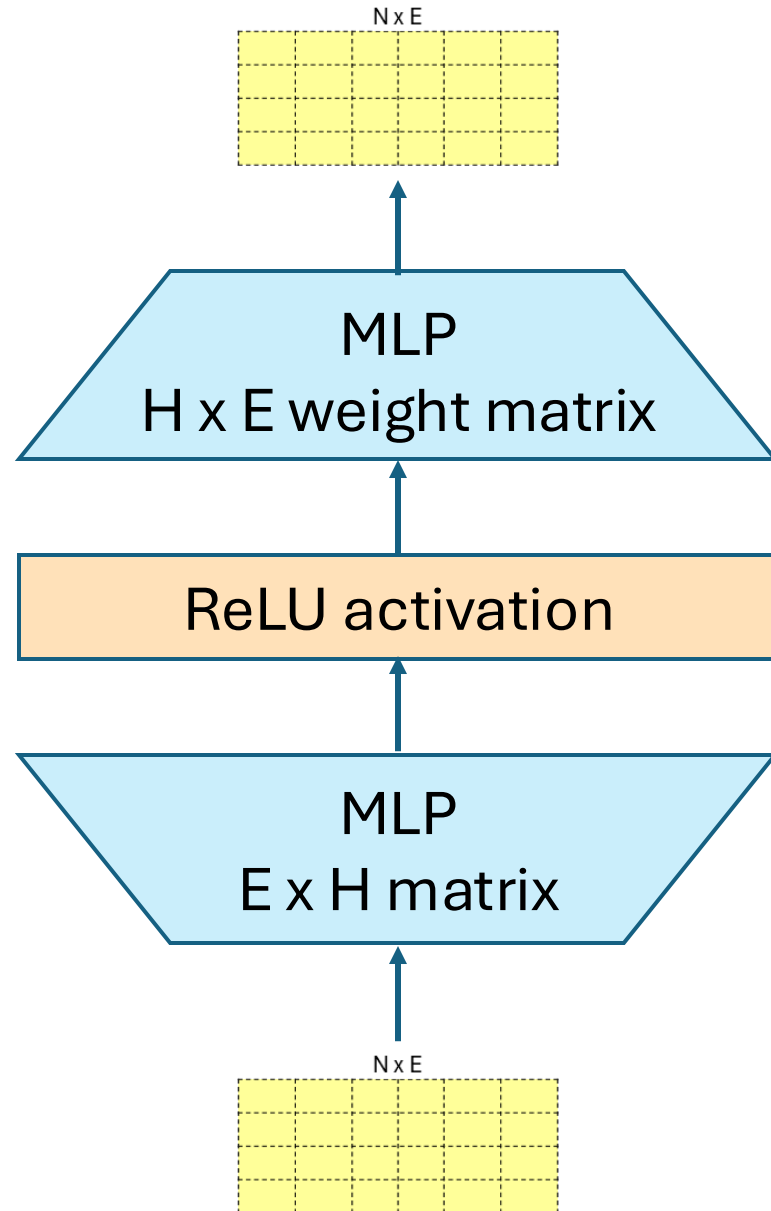
Number of parameters
 $= 4 \times (E^2 + E)$
 $\approx 4 \times E^2$



Model Size: Transformer components

Feed-Forward Network

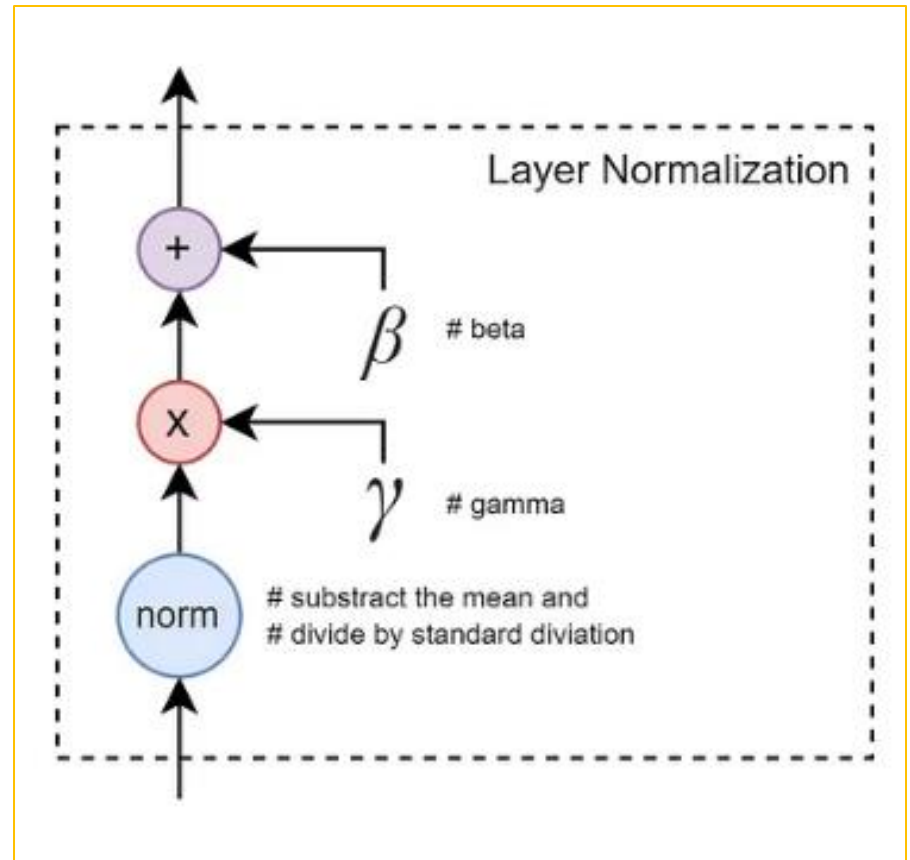
Number of parameters
 $= (E \times H + H) + (H \times E + E)$
 $= 2 \times E \times H + E + H$
 $\approx 2 \times E \times H$



Model Size: Transformer components

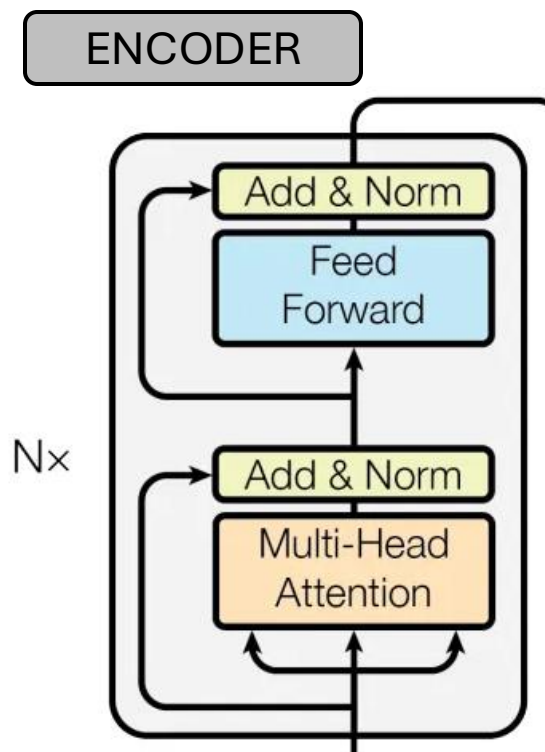
Add & Norm

Number of parameters
 $= 2 \times E \approx 0$



Model Size: Transformer components

Encoder component	Exact formula	Approximate
Multi-head attention	$4E^2 + 4E$	$4E^2$
Feed-forward network	$2EH + E + H$	$2EH$
Layer normalization	$2E$	0



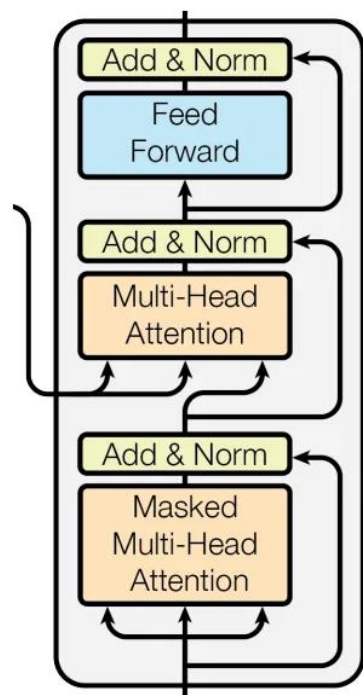
Exact: $4E^2 + 2EH + 9E + H$

Approximate: $4E^2 + 2EH$

Model Size: Transformer components

Decoder component	Exact formula	Approximate
Multi-head attention	$4E^2 + 4E$	$4E^2$
Feed-forward network	$2EH + E + H$	$2EH$
Layer normalization	$2E$	0

DECODER



Exact: $8E^2 + 2EH + 15E + H$

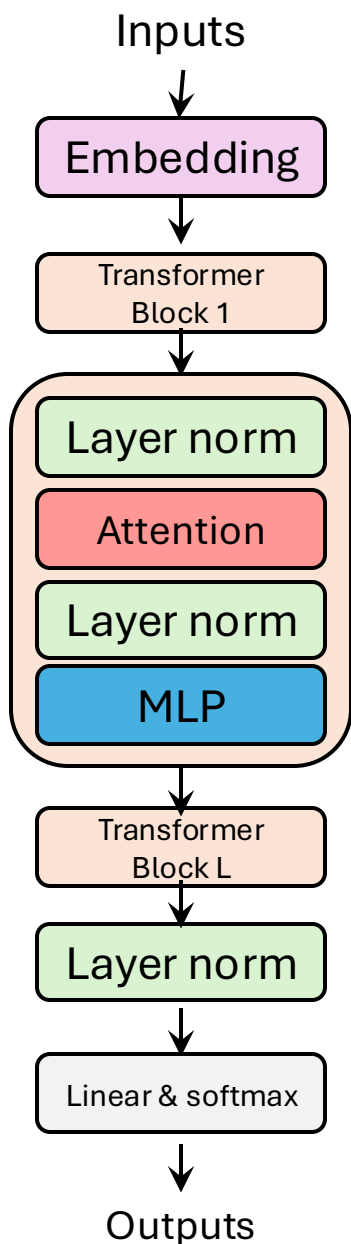
Approximate: $8E^2 + 2EH$

GPT2

<https://michaelwornow.net/2024/01/18/counting-params-in-transformer>

Overall Formula for GPT-2 Parameter Count

$E = 768$ = size of the embedding vector
 $H = 3,072$ = size of hidden layer in MLP
 $V = 50,257$ = number of tokens in vocabulary
 $P = 1,024$ = maximum sequence length
 $L = 12$ = number of transformer layers



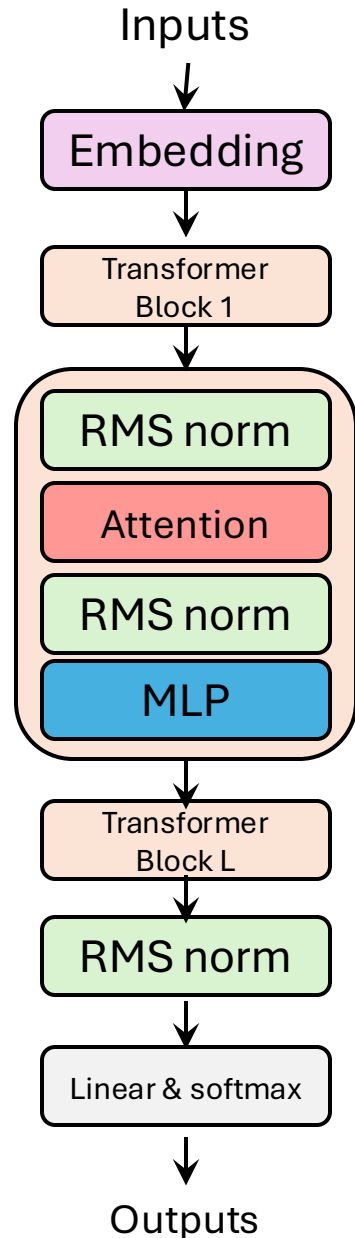
Component	Formula		GPT2
Embedding	$VE + PE$		39,383,808
• Layer norm	$2E$		1,536
• Attention	$4E^2 + 4E$	+	2,362,368
• Layer norm	$2E$	+	1,536
• MLP	$2EH + E + H$	+	4,722,432
	Sum	=	7,087,872
L layers	$L \times \text{Sum}$	$\times 12 =$	85,054,464
Layer norm	$2E$		1,536
Linear & softmax			-
TOTAL			124,439,808

LLaMA 2 and 3

S Chinni. Mastering Llama Math (Part-1): A Step-by-Step Guide to Counting Parameters in Llama-2. Medium (22 Oct 2023).

Llama-2-13B

$E = 5120$ $V = 32000$ $H = 13824$ $L = 40$



Component	Formula	Llama-2-13B
Embedding	$V \times E$	163,840,000
• RMSNorm	E	5,120
• Attention	$4 \times E^2$	+ 104,857,600
• RMSNorm	E	+ 5,120
• MLP	$3 \times E \times H$	+ 212,336,640
	Sum	= 317,204,480
L layers	$L \times \text{Sum}$	x40= 12,688,179,200
RMSNorm	E	5,120
Linear & softmax	$V \times E$	163,840,000
TOTAL		13,015,864,320

Family of Models: Llama 3.1

Category	Benchmark	Llama 3 8B	Gemma 2 9B	Mistral 7B	Llama 3 70B	Mixtral 8x22B	GPT 3.5 Turbo	Llama 3 405B	Nemotron 4 340B	GPT-4 (omni)	GPT-4o	Claude 3.5 Sonnet
General	MMLU (5-shot)	69.4	72.3	61.1	83.6	76.9	70.7	87.3	82.6	85.1	89.1	89.9
	MMLU (0-shot, CoT)	73.0	72.3 [△]	60.5	86.0	79.9	69.8	88.6	78.7 [◊]	85.4	88.7	88.3
	MMLU-Pro (5-shot, CoT)	48.3	–	36.9	66.4	56.3	49.2	73.3	62.7	64.8	74.0	77.0
	IFEval	80.4	73.6	57.6	87.5	72.7	69.9	88.6	85.1	84.3	85.6	88.0
Code	HumanEval (0-shot)	72.6	54.3	40.2	80.5	75.6	68.0	89.0	73.2	86.6	90.2	92.0
	MBPP EvalPlus (0-shot)	72.8	71.7	49.5	86.0	78.6	82.0	88.6	72.8	83.6	87.8	90.5
Math	GSM8K (8-shot, CoT)	84.5	76.7	53.2	95.1	88.2	81.6	96.8	92.3 [◊]	94.2	96.1	96.4 [◊]
	MATH (0-shot, CoT)	51.9	44.3	13.0	68.0	54.1	43.1	73.8	41.1	64.5	76.6	71.1
Reasoning	ARC Challenge (0-shot)	83.4	87.6	74.2	94.8	88.7	83.7	96.9	94.6	96.4	96.7	96.7
	GPQA (0-shot, CoT)	32.8	–	28.8	46.7	33.3	30.8	51.1	–	41.4	53.6	59.4
Tool use	BFCL	76.1	–	60.4	84.8	–	85.9	88.5	86.5	88.3	80.5	90.2
	Nexus	38.5	30.0	24.7	56.7	48.5	37.2	58.7	–	50.3	56.1	45.7
Long context	ZeroSCROLLS/QuALITY	81.0	–	–	90.5	–	–	95.2	–	95.2	90.5	90.5
	InfiniteBench/En.MC	65.1	–	–	78.2	–	–	83.4	–	72.1	82.5	–
	NIH/Multi-needle	98.8	–	–	97.5	–	–	98.1	–	100.0	100.0	90.8
Multilingual	MGSM (0-shot, CoT)	68.9	53.2	29.9	86.9	71.1	51.4	91.6	–	85.9	90.5	91.6

Table 2 Performance of finetuned Llama 3 models on key benchmark evaluations. The table compares the performance of the 8B, 70B, and 405B versions of Llama 3 with that of competing models. We **boldface** the best-performing model in each of three model-size equivalence classes. [△]Results obtained using 5-shot prompting (no CoT). [◊]Results obtained without CoT. [◊]Results obtained using zero-shot prompting.

Large Language Models - Outline

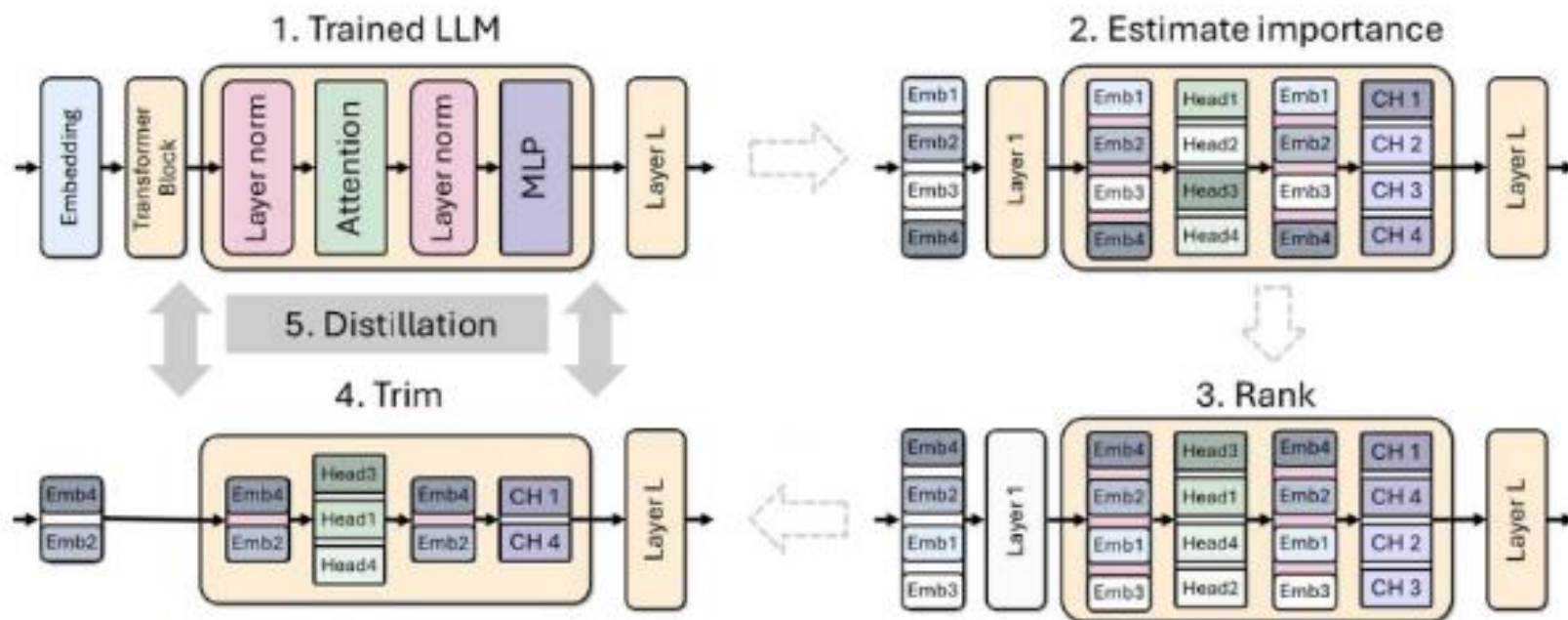
- Word Embedding
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- LLM Benchmarks (tests of intelligence)
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Family of Models and Distillation

S Sreenivas et al. *How to Prune and Distill Llama-3.1 8B to an NVIDIA Llama-3.1-Minitron 4B Model*. NVIDIA DEVELOPER (14 Aug 2024).

NVIDIA's experience: 15B model pruned and distilled to get 8B model and then 4B model

Step 1: Pruning

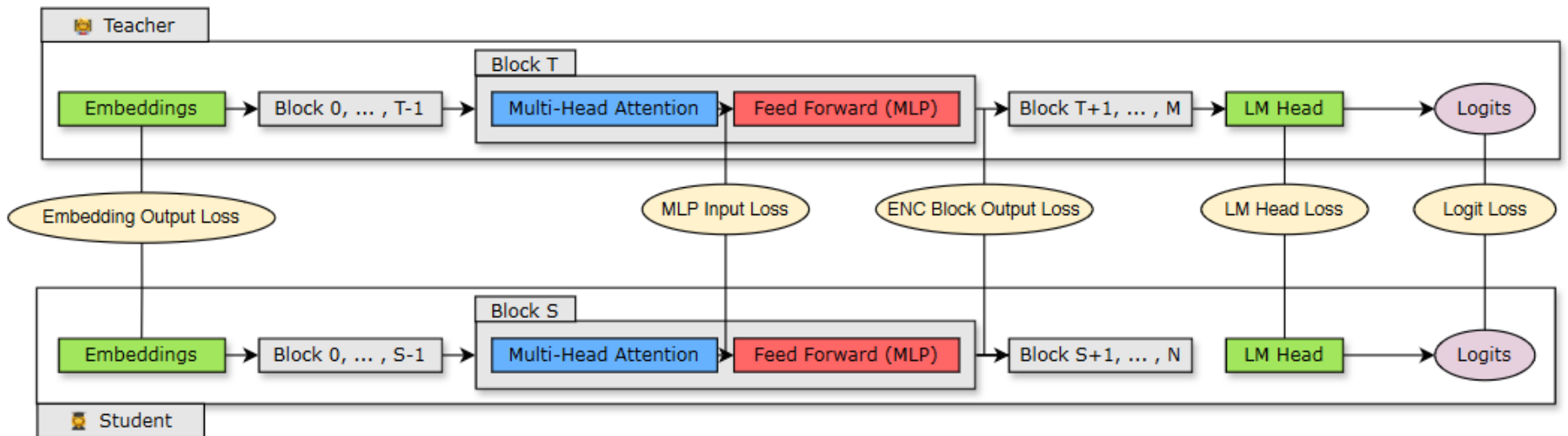


Importance analysis:

- Based on only forward propagation passes for a small calibration dataset of 1024 samples. [More straightforward than doing backpropagation and relying on gradient information.]

Family of Models and Distillation

Step 2: Distillation



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LLM Benchmarks (tests of intelligence)

Performance (according to NVIDIA)

			META	NVIDIA Llama-3.1- Minitron 4B	
Benchmark	No. of shots	Metric	Llama-3.1 8B	Depth- pruned	Width- pruned
hellaswag	10	acc_norm	0.8180	0.7321	0.7606
winogrande	5	acc	0.7727	0.7214	0.7348
MMLU	5	acc	0.6528	0.5871	0.6053
arc_challenge	25	acc_norm	0.5794	0.5256	0.5555
gsm8k	5	acc	0.4860	0.1676	0.4124
MBPP	0	pass@1	0.4227	0.3067	0.3240
Training Tokens			15T		

NVIDIA	Microsoft	Google	Alibab
Minitron 4B	Phi-2 2.7B	Gemma 2 2.6B	Qwen 2 1.5B
0.7496	0.7524	0.7300	0.6660
0.7403	0.7400	0.7090	0.6620
0.5860	0.5749	0.5130	0.5650
0.5085	0.6100	0.5540	0.4390
0.2411	0.5500	0.2390	0.5850
0.2817	0.4700	0.2900	0.3740
94B	1.4T	3T	7T

Observations:

- Width-only pruning (change H and E) is better than depth-only pruning (cutting layers)
- As expected, Llama-3.1-Minitron will not perform as well as the Llama-3.1-8B model
- Smaller models can perform better than the Llama-3.1-8B model for specific tasks

LLM Benchmarks (tests of intelligence)

HellaSwag (Zellers et al, 2018)

- Humans: 95%+ accuracy

A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She...

- A. rinses the bucket off with soap and blow dry the dog's head.
- B. uses a hose to keep it from getting soapy.
- C. gets the dog wet, then it runs away again.**
- D. gets into a bath tub with the dog.

Come to a complete halt at a stop sign or red light. At a stop sign, come to a complete halt for about 2 seconds or until vehicles that arrived before you clear the intersection. If you're stopped at a red light, proceed when the light has turned green. ...

- A. Stop for no more than two seconds, or until the light turns yellow. A red light in front of you indicates that you should stop.
- B. After you come to a complete stop, turn off your turn signal. Allow vehicles to move in different directions before moving onto the sidewalk.
- C. Stay out of the oncoming traffic. People coming in from behind may elect to stay left or right.
- D. If the intersection has a white stripe in your lane, stop before this line. Wait until all traffic has cleared before crossing the intersection.**

LLM Benchmarks (tests of intelligence)

WinoGrande

- A large-scale dataset of 44k problems, inspired by Winograd Schema Challenge (Levesque, Davis and Morgenstern, 2011)
- Benchmark for commonsense reasoning
- Pronoun resolution problem
- Human: 94% accuracy

Twin sentences		Options (answer)
The monkey loved to play with the balls but ignored the blocks because he found them exciting.	The monkey loved to play with the balls but ignored the blocks because he found them dull.	balls / blocks
William could only climb beginner walls while Jason climbed advanced ones because he was very weak.	William could only climb beginner walls while Jason climbed advanced ones because he was very strong.	William / Jason
Robert woke up at 9:00am while Samuel woke up at 6:00am, so he had less time to get ready for school.	Robert woke up at 9:00am while Samuel woke up at 6:00am, so he had more time to get ready for school.	Robert / Samuel
The child was screaming after the baby bottle and toy fell. Since the child was hungry, it stopped his crying.	The child was screaming after the baby bottle and toy fell. Since the child was full, it stopped his crying.	baby bottle / toy

LLM Benchmarks (tests of intelligence)

MMLU (Measuring Massive Multitask Language Understanding)

- 16k multiple-choice questions spanning 57 academic subjects including mathematics, philosophy, law and medicine (Hendrycks et al, 2020)
- Human expert: estimated 89.8% accuracy

Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.

(A) 0 (B) 1 (C) 2 (D) 3

Would a reservation to the definition of torture in the ICCPR be acceptable in contemporary practice?

(A) This is an acceptable reservation if the reserving country's legislation employs a different definition

(B) This is an unacceptable reservation because it contravenes the object and purpose of the ICCPR

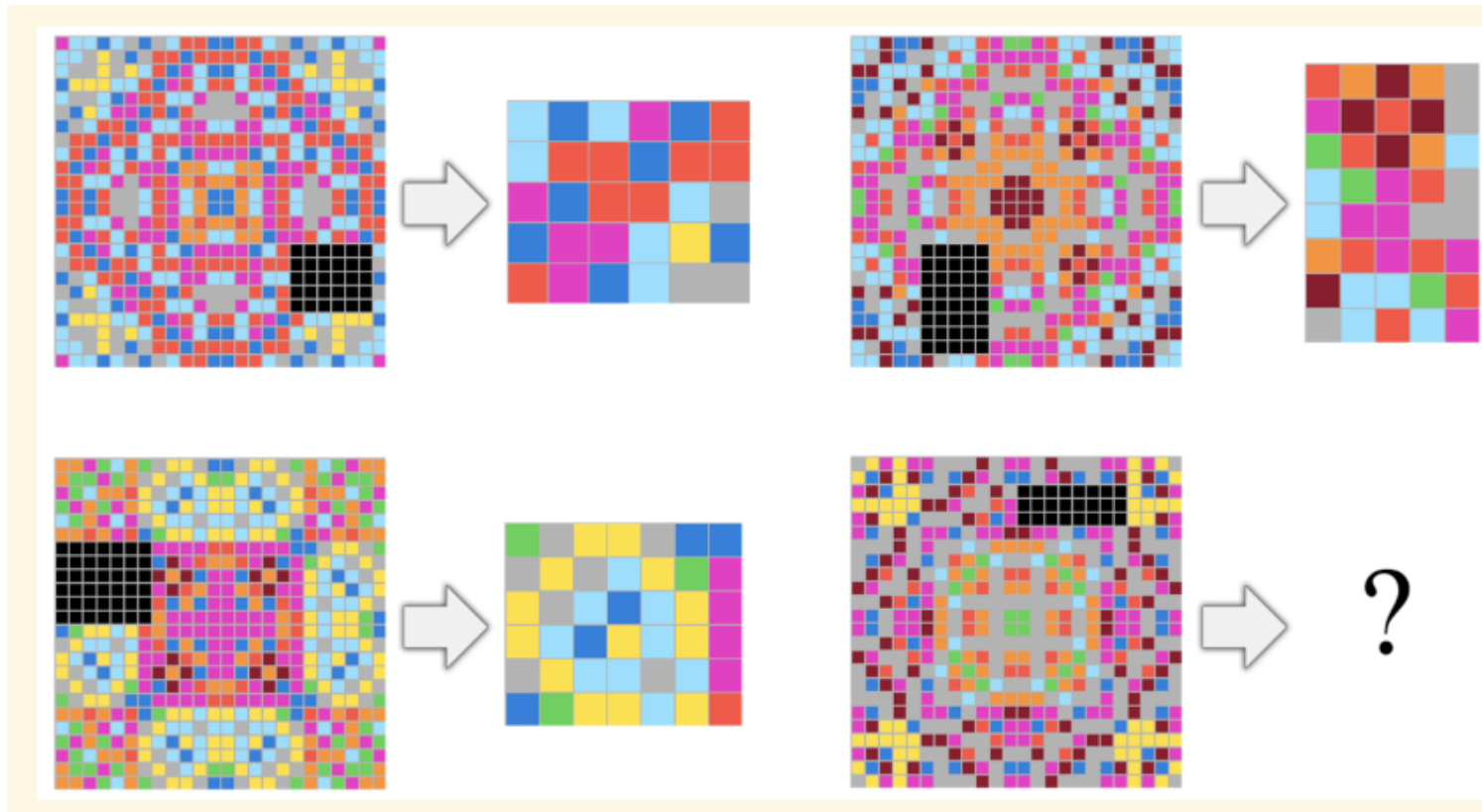
(C) This is an unacceptable reservation because the definition of torture in the ICCPR is consistent with customary international law

(D) This is an acceptable reservation because under general international law States have the right to enter reservations to treaties

LLM Benchmarks (tests of intelligence)

ARC (Abstraction and Reasoning Corpus) **Challenge** (Chollet 2019)

- Humans don't need training
- Assumed that humans would score close to 100%



LLM Benchmarks (tests of intelligence)

GSM8K

- a dataset of 8.5K
- grade school math word problems: solutions use basic arithmetic operations (+ − × ÷)
- multi-step mathematical reasoning: takes between 2 and 8 steps to solve

Problem: Beth bakes 4, 2 dozen batches of cookies in a week. If these cookies are shared amongst 16 people equally, how many cookies does each person consume?

Solution: Beth bakes 4 2 dozen batches of cookies for a total of $4 \times 2 = 8$ dozen cookies
There are 12 cookies in a dozen and she makes 8 dozen cookies for a total of $12 \times 8 = 96$ cookies
She splits the 96 cookies equally amongst 16 people so they each eat $96/16 = 6$ cookies

Final Answer: 6

Problem: Mrs. Lim milks her cows twice a day. Yesterday morning, she got 68 gallons of milk and in the evening, she got 82 gallons. This morning, she got 18 gallons fewer than she had yesterday morning. After selling some gallons of milk in the afternoon, Mrs. Lim has only 24 gallons left. How much was her revenue for the milk if each gallon costs \$3.50?

Mrs. Lim got 68 gallons - 18 gallons = 50 gallons this morning.

So she was able to get a total of 68 gallons + 82 gallons + 50 gallons = 200 gallons.

She was able to sell 200 gallons - 24 gallons = 176 gallons.

Thus, her total revenue for the milk is $\$3.50/\text{gallon} \times 176 \text{ gallons} = \616 .

Final Answer: 616

LLM Benchmarks (tests of intelligence)

MBPP (Mostly Basic Python Programming) (Austin et al, 2021)

- ~1,000 Python programming tasks solvable by entry-level programmers
- Each problem: natural language task description, code solution, 3 automated test cases

prompt

Write a function to find the smallest missing element in a sorted array. Your code should satisfy these tests:

```
assert smallest_missing([0, 1, 2, 3, 4, 5, 6], 0, 6) == 7
```

```
assert smallest_missing([0, 1, 2, 6, 9, 11, 15], 0, 6) == 3
```

```
assert smallest_missing([1, 2, 3, 4, 6, 9, 11, 15], 0, 7) == 0
```

model

```
def smallest_missing(arr, n, m):
```

```
    smallest = min(n, m)
```

```
    for i in range(n, m + 1):
```

```
        if arr[i] <= smallest:
```

```
            smallest += 1
```

```
    return smallest
```

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Low Rank Adaptation (LoRA)

LORA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS,

Ed Hu, Y. Shen, P. Wallis, Z. Zhu, Y. Li, S. Wang, L. Wang, W. Chen (Microsoft), Oct 2021

<https://arxiv.org/pdf/2106.09685.pdf#page=5>

- Paper has 11k+ citations
- Ground-breaking and efficient fine-tuning technique
- LoRA's major application is to finetune large AI models with low compute

Fine-tuning

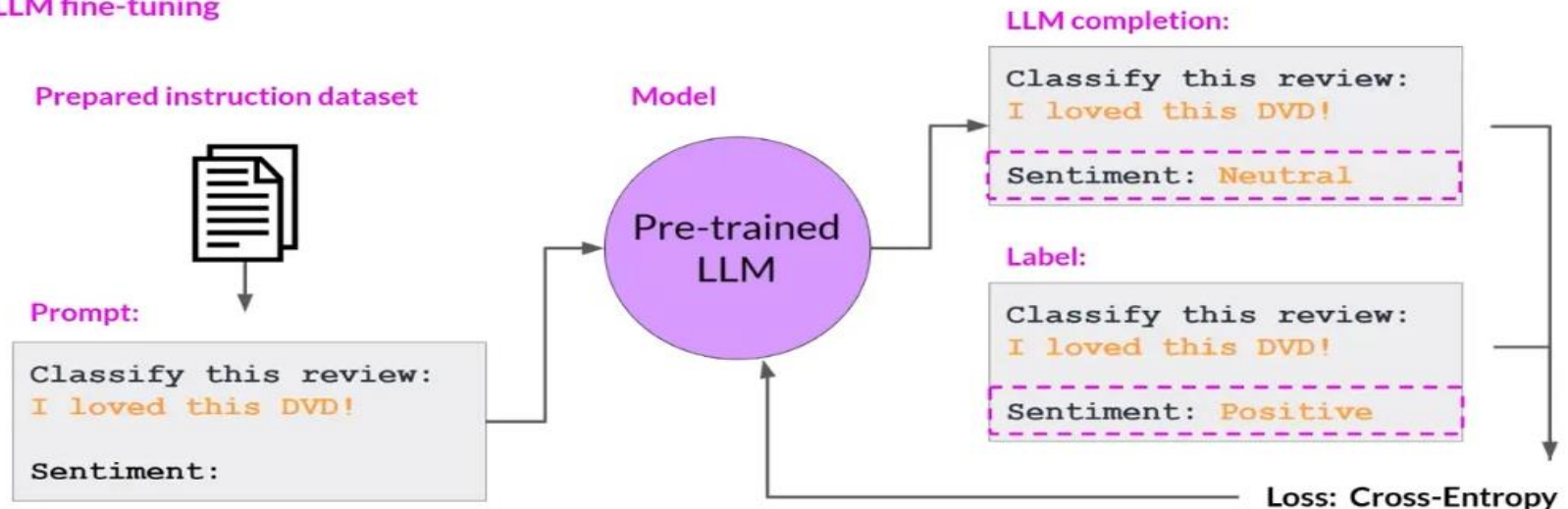
Approach depend on amount of data collected

- Lot – training from scratch
- **Medium – fine tuning**
- Sparse – use the general model

What is fine-tuning (aka transfer learning)?

- updates the parameters of a pre-trained model with new data.
- adapts to characteristics of target domain/task while preserving general knowledge learned from large-scale pre-training, e.g., struggle to respond to specific prompts or instructions

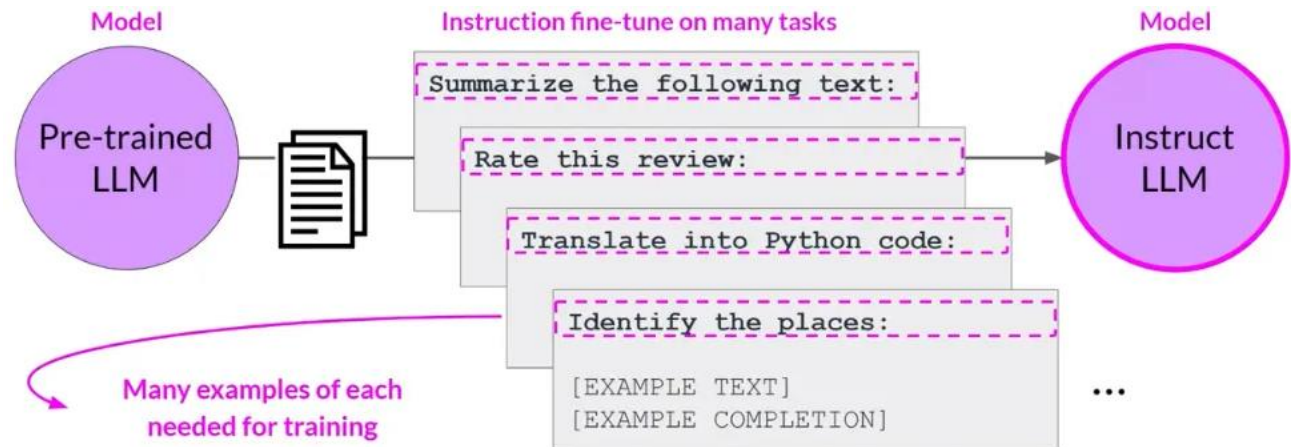
LLM fine-tuning



Fine-tuning – Challenges and Solutions

Catastrophic Forgetting – model forgets previously learned information when trained on new data, i.e., improved performance on targeted task but degradation with other tasks.

Possible Solution:
fine-tuning across
a wide range of
instruction types



Different Fine-tuning Approaches

- Finetune all parameters or last k layers (say k=2) or bias-only
- Adapter tuning – insert 2 FC layers (at end), increase latency

Problems with Fine-tuning

- retraining all the model parameters (billions for LLMs) requires vast computational resources, memory, time, energy and money
- overfitting, with small new data, degrade performance on other tasks.

LoRA: Background concepts

Rank of a matrix is the dimension of the vector space generated by its columns (rows) i.e. no. of linearly independent columns (or rows).

- Matrix W_{nk} (n rows, k columns), $rank(W) \leq \min(n, k)$

$$rank \left(\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 0 \\ 3 & 0 & 4 \end{bmatrix} \right) = 3 \qquad rank \left(\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \right) = 1$$

- **Low-rank matrix** W_{nk} if $rank(W) \ll \min(n, k)$.
- Rank decomposition matrix $W_{nk} = B_{nr} A_{rk}$ where $rank(W) = r$

Hypothesis: W_{nk} = weights of a single layer

ΔW_{nk} (due to fine-tuning) is a low-rank matrix

Thus $\Delta W_{nk} = B_{nr} A_{rk}$ where $r \ll \min(n, k)$

LoRA based on low-rank approximation, which reduces complexity and dimensionality of large matrices by finding smaller matrices that approximate them well.

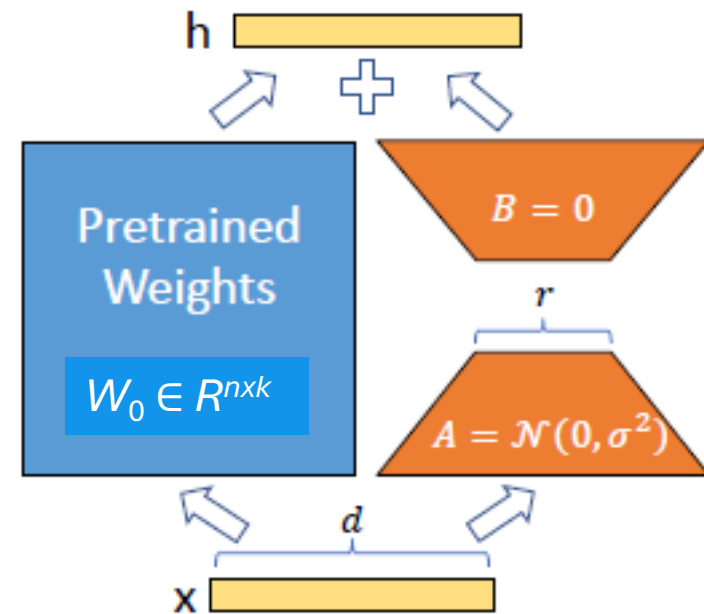
LoRA: Main idea

- For a **pre-trained weight matrix** W_0 , its update $\Delta W_0 \in R^{n \times k}$ is represented with a **low-rank decomposition** BA :

$$W_0 + \Delta W_0 = W_0 + B_{nr} A_{rk}$$

- During training, W_0 is **frozen** and does not receive gradient updates. where A and B contain **trainable parameters**.
- A random Gaussian initialization is used for A and zero is used for B , so $\Delta W_0 = BA$ is zero at the beginning of training.
- When deployed in production, we can explicitly compute and store $W_0 + BA$ and perform inference.
- For $h = W_0 x$, the **modified forward pass** yields:

$$h = W_0 x + \Delta W_0 x = W_0 x + B_{nr} A_{rk} x$$



LoRA: Advantages

- Reduces no. of trainable parameters by orders of magnitude

$$\left(\begin{array}{c} \boxed{W} \\ \text{frozen} \end{array} + \begin{array}{c} \boxed{B} \boxed{A} \\ \text{trainable} \end{array} \right) x = h(x)$$

W is $n \times k$ matrix
 B is $n \times r$ matrix
 A is $r \times k$ matrix
 x is $k \times 1$ matrix
 $h(x)$ is $n \times 1$ matrix

Example: $n \times k = 512 \times 64 = 32768$, $r = 8$ (suggested 4-32)
 $\Rightarrow n \times r + r \times k = 512 \times 8 + 8 \times 64 = 4096 + 512$ (86% reduction)

- r is usually very small and can be adjusted to get best result
- Reduction of training time, GPU memory size and cost
- Avoid overfitting by keeping original weights fixed and updating a small fraction of weights only, preserving generalization ability of model while adapting to new data
- No additional inference time (no appending layers)
- Easier task switching - storing one copy of original weights and multiple sets of update matrices for different tasks (i.e. **multiple adapters**)

LoRA: Example of usefulness

Efficient Fine-Tuning for Meta's Llama-v2-7B on a Single GPU

Motivation – memory bottleneck (table below not most up-to-date)

GPU	H100	A100	V100	RTX4090	A10G	T4	RTX4080
VRAM (GB)	80	80/40	32/16	48	24	16	16

Expensive and in shortage

Memory requirement for storing the 7B model:

Using **FP32** for each parameter $w \Rightarrow 7 \times 4 \text{ GB} = 28 \text{ GB}$

For training, you need the gradient for each w

Using **FP32** for each gradient $\nabla w \Rightarrow 7 \times 4 \text{ GB} = 28 \text{ GB}$

Optimizer states:

$$\text{Adam } w \leftarrow w - \alpha / (\sqrt{v} + \epsilon) m$$

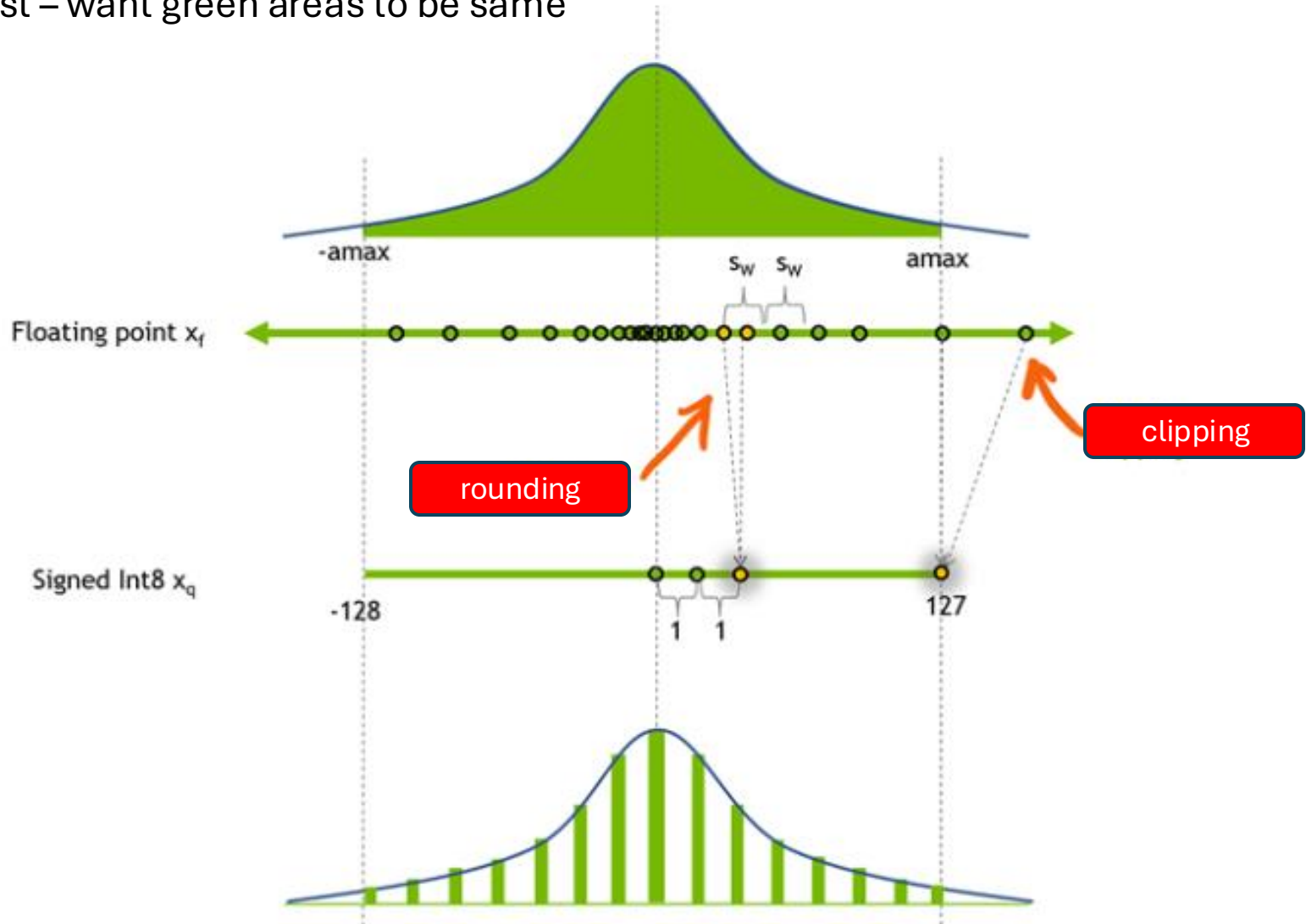
(m = momentum, v = variance)

Using **FP32** for m and v for each $w \Rightarrow 2 \times 7 \times 4 \text{ GB} = 56 \text{ GB}$

112 GB

Quantization to 8-bits (integer)

- rounding and clipping errors
- not the best – want green areas to be same



LoRA and Quantization: impact

GPU	H100	A100	V100	RTX4090	A10G	T4	RTX4080
VRAM (GB)	80	80/40	32/16	48	24	16	16

Fun fact: using LoRA on GPT-3 175B reduces no. of trainable parameters from 175 billion to 17 million, a 10,000 times reduction

Memory requirement for storing the 7B model:

Using **INT8** for each parameter $w \Rightarrow 7 \times 1 \text{ GB} = 7 \text{ GB}$

For training, you need the gradient for each w

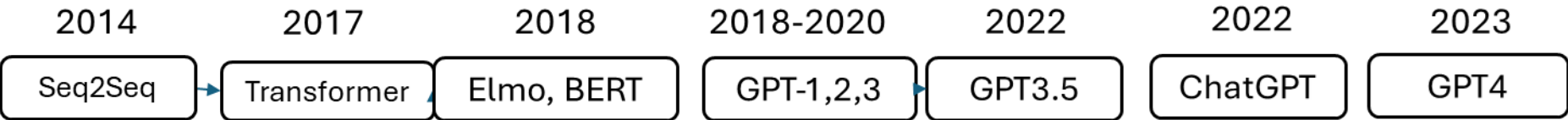
Using **FP16** for each gradient $\nabla w \Rightarrow 1 \times 2 \text{ GB} = 2 \text{ GB}$

Using **FP16** for m and v for each $w \Rightarrow 2 \times 1 \times 2 \text{ GB} = 4 \text{ GB}$

13 GB

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Hangzhou DeepSeek Artificial Intelligence Basic Technology Research Co., Ltd.



deepseek

Native name	杭州深度求索人工智能基础技术研究有限公司
Company type	Private
Industry	Information technology Artificial intelligence
Founded	May 2023
Founder	Liang Wenfeng
Headquarters	Hangzhou, Zhejiang, China
Key people	Liang Wenfeng (CEO)
Owner	High-Flyer
Number of employees	Under 200
Website	deepseek.com

Feb 2016:	High-Flyer co-founded by Liang Wenfeng
2019-2021:	High-Flyer developed as a hedge fund using AI in trading
Apr/May 2023:	High-Flyer started DeepSeek for AGI
Nov 2023:	DeepSeek-Coder
Nov 2023:	DeepSeek-LLM (7B and 67B) and chat versions
Jan 2024:	DeepSeek-MoE (16B)
Jan 2024:	DeepSeek-MoE paper
Apr 2024:	DeepSeek-Math
Apr 2024:	Group Relative Policy Optimization (GRPO) paper
May 2024:	DeepSeek-V2
Sep 2024:	DeepSeek-V2.5
Dec 2024:	DeepSeek-V3
Jan 2025:	DeepSeek-R1

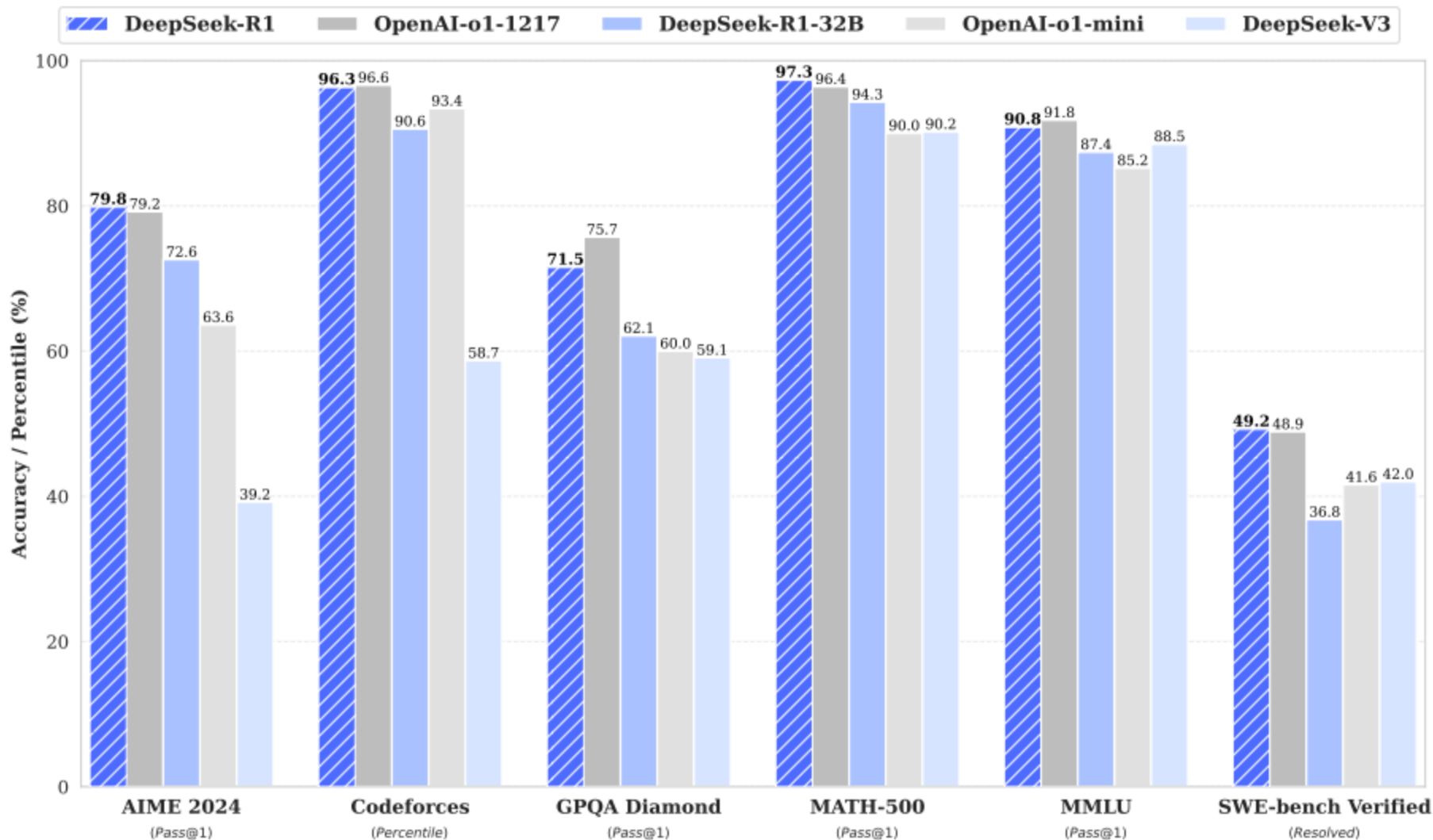


Figure 1 | Benchmark performance of DeepSeek-R1.

2024 AIME I Problems/Problem 1

Contents [hide]

1 Problem

2 Solution 1

3 Solution 2

4 Video Solution

5 See also

Problem

Every morning Aya goes for a 9-kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s + 2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s + \frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.

Solution 1

$$\frac{9}{s} + t = 4 \text{ in hours and } \frac{9}{s+2} + t = 2.4 \text{ in hours.}$$

Subtracting the second equation from the first, we get,

$$\frac{9}{s} - \frac{9}{s+2} = 1.6$$

Multiplying by $(s)(s+2)$, we get

$$9s + 18 - 9s = 18 = 1.6s^2 + 3.2s$$

Multiplying by 5/2 on both sides, we get

$$0 = 4s^2 + 8s - 45$$

Factoring gives us

$$(2s - 5)(2s + 9) = 0, \text{ of which the solution we want is } s = 2.5.$$

Substituting this back to the first equation, we can find that $t = 0.4$ hours.

Lastly, $s + \frac{1}{2} = 3$ kilometers per hour, so

$$\frac{9}{3} + 0.4 = 3.4 \text{ hours, or } \boxed{204} \text{ minutes}$$

Split (1)
test · 500 rows

Search this dataset

problem string · lengths	solution string · lengths	answer string · lengths	subject string · classes	level int64	unique_id string · lengths
					
Convert the point $(0,3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form...	We have that $r = \sqrt{0^2 + 3^2} = 3$. Also, if we draw the line connecting the origin and $(0,3)$, this...	$\left(3, \frac{\pi}{2} \right)$	Precalculus		test/precaculus/807.json
Define $[p = \sum_{k=1}^{\infty} \frac{1}{k^2}]$ and $q = \sum_{k=1}^{\infty} \frac{1}{k}$.	We count the number of times $\frac{1}{n^3}$ appears in the sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty}$.	$p - q$	Intermediate Algebra		test/intermediate_algebra/1994.json
If $f(x) = \frac{3x-2}{x-2}$, what is the value of $f(-2) + f(-1) + f(0)$? Express your answer as a common...	$f(-2) + f(-1) + f(0) = \frac{3(-2)-2}{-2-2} + \frac{3(-1)-2}{-1-2} + \frac{3(0)-2}{0-2} = \frac{-8}{-4} + \frac{-5}{-3} + \frac{-2}{-2} = 2 + \frac{5}{3} + 1 = \frac{14}{3}$	$\frac{14}{3}$	Algebra		test/algebra/2584.json
How many positive whole-number divisors does 196 have?	First prime factorize $196 = 2^2 \cdot 7^2$. The prime factorization of any divisor of 196 cannot include any...	9	Number Theory		test/number_theory/572.json
The results of a cross-country team's training run are graphed below. Which student has the greatest average...	Evelyn covered more distance in less time than Briana, Debra and Angela, so her average speed is greater than...	Evelyn	Algebra		test/algebra/1349.json
A regular hexagon can be divided into six equilateral triangles. If the perimeter of one of the triangles is...	The side length of the hexagon is equal to the side length of one of the equilateral triangles. Since the...	42	Prealgebra		test/prealgebra/1622.json
What is the smallest positive perfect cube that can be written as the sum of three consecutive integers?	The sum of three consecutive integers takes the form $(k-1) + k + (k+1) = 3k$ and hence is a multiple of 3...	27	Number Theory		test/number_theory/515.json
The set of points (x,y,z) that satisfy $\begin{bmatrix} 2x \\ 3y \\ -z \end{bmatrix}$ is a line. The set of points (x,y,z) that satisfy...	For the first line, let $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ 3t \\ -t \end{bmatrix}$. Then $\begin{bmatrix} 2x \\ 3y \\ -z \end{bmatrix} = \begin{bmatrix} 4t \\ 9t \\ t \end{bmatrix}$.	9π	Precalculus		test/precaculus/927.json
What is the distance, in units, between the points $(2, -6)$ and $(-4, 3)$? Express your answer in simplest...	We use the distance formula: $\sqrt{(2 - (-4))^2 + ((-6) - 3)^2} = \sqrt{6^2 + (-9)^2} = \sqrt{36 + 81} = \sqrt{117} = 3\sqrt{13}$	$3\sqrt{13}$	Algebra		test/algebra/2036.json
The expression $2(3 \cdot 4) + 5$ is equal to 121, since multiplication is carried out before...	By the associative property of multiplication, it doesn't help to insert parentheses that specify the...	4	Prealgebra		test/prealgebra/1139.json
What is the least positive integer multiple of 30 that can be written with only the digits 0 and 2?	Let M be the least positive multiple of 30 that can be written with only the digits 0 and 2. First, M is...	2220	Number Theory		test/number_theory/1032.json
Let $p(x)$ be a polynomial of degree 5 such that $p(n) = \frac{n^2 - 1}{n}$ for $n = 2, 3, 4, \dots, 7$.	Let $q(x) = (x^2 - 1)p(x) - x$. Then $q(x)$ has degree 7, and $q(n) = 0$ for $n = 2, 3, 4, \dots, 7$, so $q(x) = (x-2)(x-3)(x-4)(x-5)(x-6)(x-7)(x-a)$.	$\frac{1}{66}$	Intermediate Algebra		test/intermediate_algebra/1197.json
The proper divisors of 12 are 1, 2, 3, 4 and 6. A proper divisor of an integer N is a positive divisor...	Prime factorize $284 = 2^2 \cdot 71$. The sum of the proper divisors of 284 is $1 + 2 + 2^2 + 71 + 2 \cdot 71 = 146$.	284	Number Theory		test/number_theory/737.json
The volume of the cylinder shown is 45π cubic cm. What is the height in centimeters of the cylinder?	The volume of the cylinder is $\pi r^2 h$. The radius of the base is 3 cm, so we have $9\pi h = 45\pi$.	5	Geometry		test/geometry/248.json

LLM Training Process (in general)



Pre-training

- Start with an architecture
- Pre-train on a big dataset
- Goal: to be good at next token/word prediction
- ▶ *Costly when architecture has lots of parameters and dataset is huge*
- ▶ *Not reliable at following instructions: prompt does not trigger what you want*

Supervised fine-tuning (SFT)

- Teach the LLM to follow instructions
- Train on dataset of (instruction, response)
- ▶ *May want to use LoRA*

Reinforcement learning (RL)

- Teach the LLM to do better with feedback:
 - Human feedback (HF)
 - AI feedback
- ▶ *Human feedback can be costly*

DeepSeek-V3 (the chat model)

Pre-training

- Start with 671B parameter **MoE** architecture
- Trained on 14.8T tokens: multilingual corpus, mostly English and Chinese and higher ratio of math and coding data
- Extend context length from 4K to 32K to 128K, using YaRN
- Resulting in **DeepSeek-V3-Base** (starting point for DeepSeek-R1-Zero)
- Cost: US\$5.328m

Supervised fine-tuning (SFT)

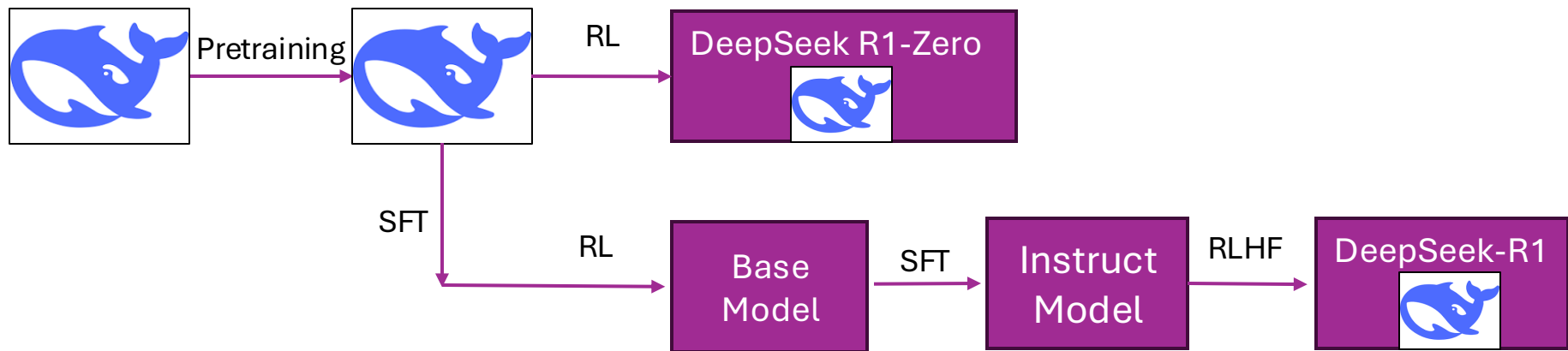
- Trained expert models to generate samples of reasoning data (math, programming, logic) and used V2.5 to generate non-reasoning data
- Trained DeepSeek-V3-Base with the 1.5M samples for 2 epochs
- Resulting in “SFT checkpoint of V3”

Reinforcement learning (RL)

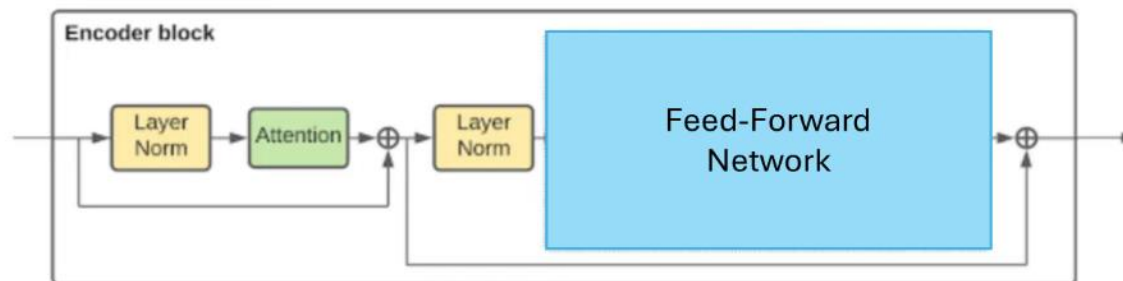
- Created reward model starting from SFT checkpoint of V3
- Train SFT checkpoint of V3 with **GRPO** using both reward model and **rule-based reward**
- Resulting in DeepSeek-V3

Total training cost: < US\$5.6m

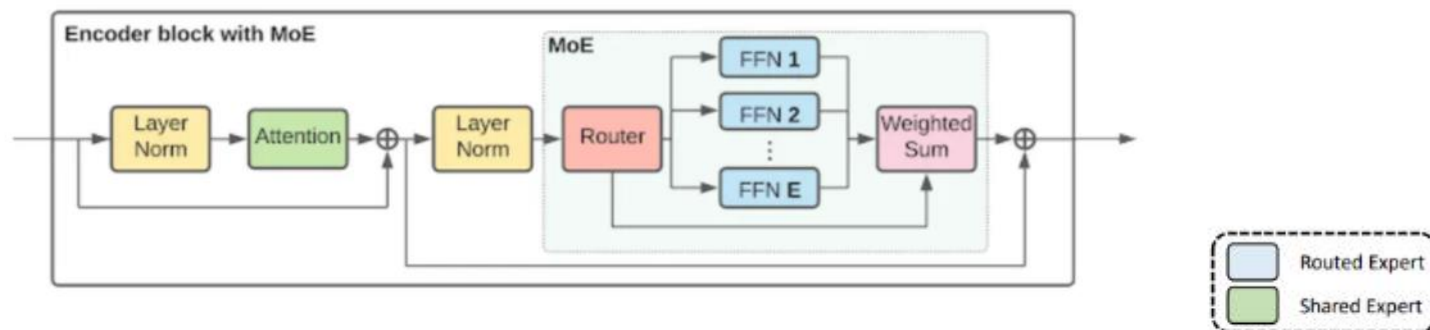
Training pipeline



Example of
FFN in
transformer



Mixture-of-
Experts
(MoE)
replaces FFN



Explanation of
DeepSeek's MoE
architecture:

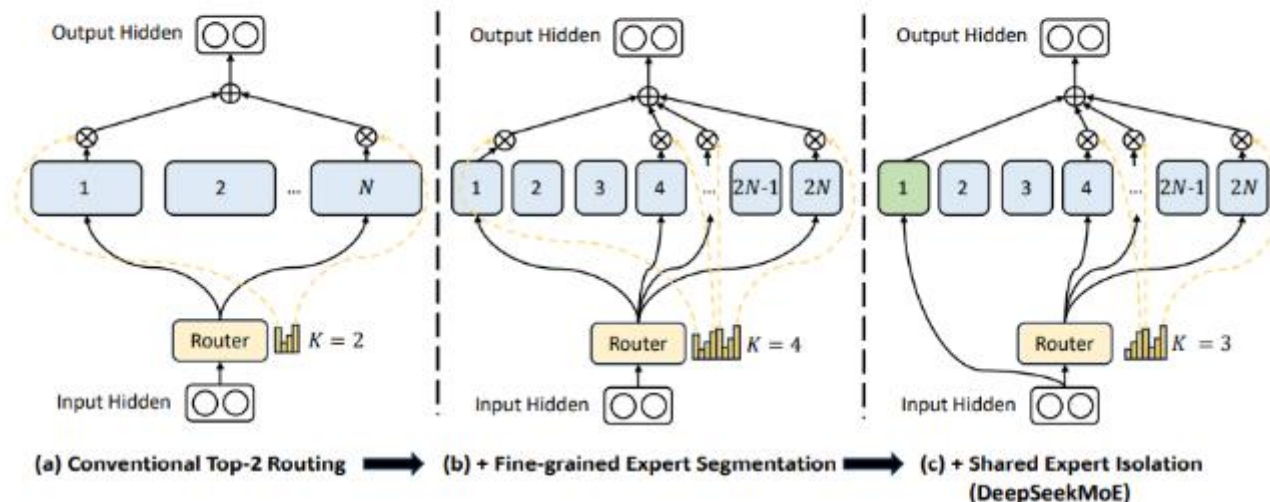


Figure 2 | Illustration of DeepSeekMoE. Subfigure (a) showcases an MoE layer with the conventional top-2 routing strategy. Subfigure (b) illustrates the fine-grained expert segmentation strategy. Subsequently, subfigure (c) demonstrates the integration of the shared expert isolation strategy, constituting the complete DeepSeekMoE architecture. It is noteworthy that across these three architectures, the number of expert parameters and computational costs remain constant.

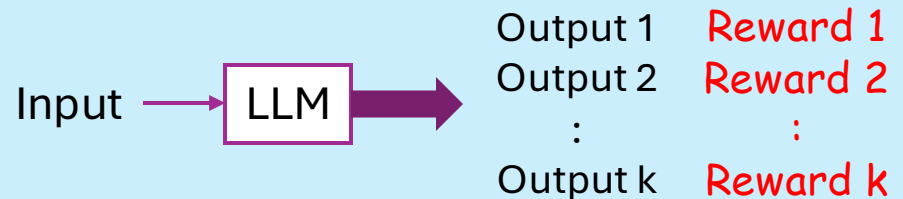
DeepSeek-R1-Zero (analogy with Alpha Go Zero – pure RL)

Pre-training

- Start with 671B parameter DeepSeek-V3-Base which was pretrained on 14.8T tokens: mostly English and Chinese and higher ratio of math and coding data

Reinforcement learning (RL)

- Group Relative Policy Optimization (GRPO)

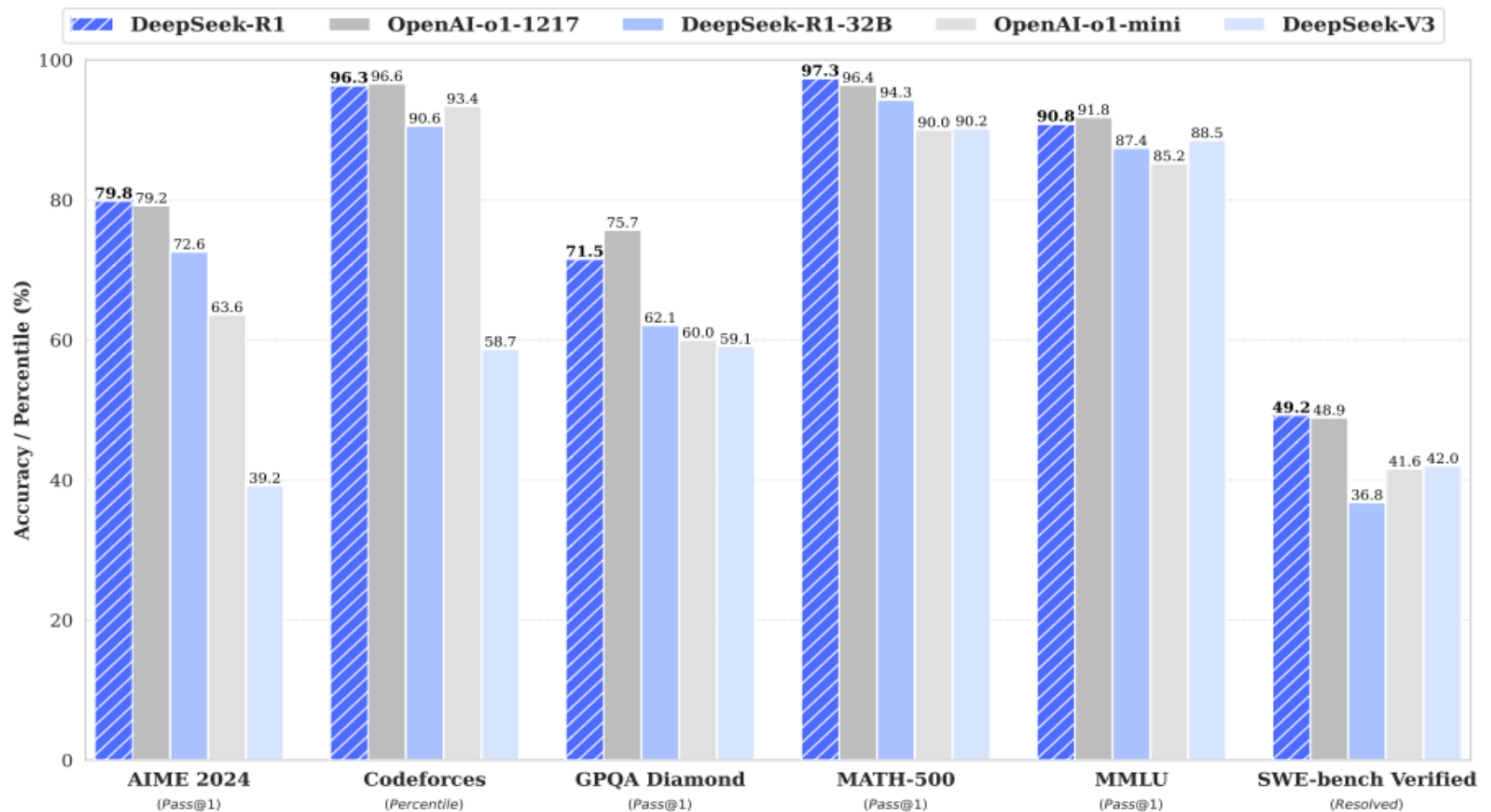


- Rule-based reward
 - Accuracy
 - Format

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>` `</think>` and `<answer>` `</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: **prompt**. Assistant:

Table 1 | Template for DeepSeek-R1-Zero. **prompt** will be replaced with the specific reasoning question during training.

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444



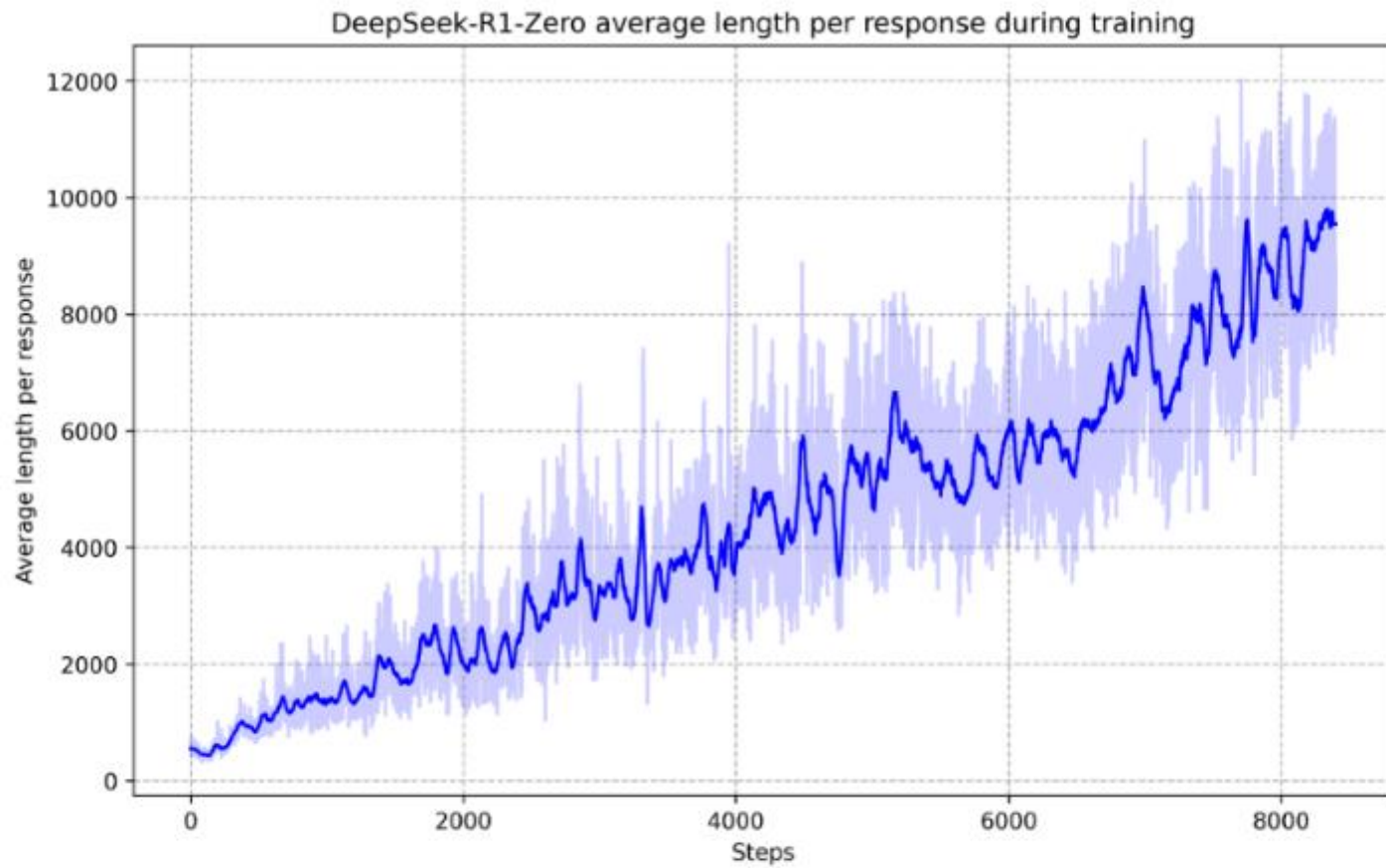


Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\dots$

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

$\dots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\dots$

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: $\dots$

$\dots$

Table 3 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-R1 (to solve (a) readability issues (b) language mixing problem)

[1] SFT

- Start with DeepSeek-V3-Base
- Train on high-quality readable results from DeepSeek-R1-Zero

Fine-tune for readability

[2] RL for reasoning

- GRPO on reasoning problems (math, programming, logic) with rule-based rewards

Reinforcing good reasoning patterns

[3] SFT

- Use model from [2] to generate lots of reasoning samples and use DeepSeek-V3 to retain only correct and readable output
- Model from [2] is fine-tune trained on retained samples and some data from DeepSeek-V3's training data so that the model is exposed to non-reasoning data

Fine-tune for good reasoning and non-reasoning capabilities

[4] RL for diverse tasks

- GRPO on diverse tasks with rule-based rewards used where appropriate and LLM feedback to align with human preferences otherwise

Reinforce for all scenarios

Benchmark (Metric)		Claude-3.5- Sonnet-1022	GPT-4o 0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
Architecture		-	-	MoE	-	-	MoE
# Activated Params		-	-	37B	-	-	37B
# Total Params		-	-	671B	-	-	671B
English	MMLU (Pass@1)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2
	IF-Eval (Prompt Strict)	86.5	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	75.7	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	47.0	30.1
	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	82.5
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	87.6
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	92.3
Code	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	65.9
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	96.6	96.3
	Codeforces (Rating)	717	759	1134	1820	2061	2029
	SWE Verified (Resolved)	50.8	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	61.7	53.3
Math	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	79.8
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	97.3
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	78.8
Chinese	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	92.8
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	91.8
	C-SimpleQA (Correct)	55.4	58.7	68.0	40.3	-	63.7

Table 4 | Comparison between DeepSeek-R1 and other representative models.

Distillation of Deep-Seek-R1

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Table 5 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks.

For the distillation, SFT only on 800k samples

DeepSeek-R1-Zero

Pre-Training → ~~Supervised Fine-Tuning~~ → Reinforcement Learning

- We directly apply RL to the base model without relying on supervised fine-tuning (SFT) as a preliminary step. This approach allows the model to explore chain-of-thought (CoT) for solving complex problems, resulting in the development of DeepSeek-R1-Zero. DeepSeek-R1-Zero demonstrates capabilities such as self-verification, reflection, and generating long CoTs, marking a significant milestone for the research community. Notably, it is the first open research to validate that reasoning capabilities of LLMs can be incentivized purely through RL, without the need for SFT. This breakthrough paves the way for future advancements in this area.

DeepSeek-R1

Pre-Training → Supervised Fine-Tuning → Reinforcement Learning → Supervised Fine-Tuning → Reinforcement Learning

- We introduce our pipeline to develop DeepSeek-R1. The pipeline incorporates two RL stages aimed at discovering improved reasoning patterns and aligning with human preferences, as well as two SFT stages that serve as the seed for the model's reasoning and non-reasoning capabilities. We believe the pipeline will benefit the industry by creating better models.